

The Cost of Charitable Care: Resource Externalities and the Value of Insurance*

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Abstract

Charitable care provides free and discounted hospital access for millions of uninsured Americans. Using cross-state spillovers from insurance access expansions, we estimate charity care's costs and document an excess crowd-out of insured patients. A model of competitive hospital resource allocations reveals that limits on discrimination in emergency care induce hospitals facing greater charitable demand to reallocate resources to scheduled care. Compensating hospitals for charitable care raises patient surplus, with most gains accruing to the insured. Patient-optimal hospital compensation is zero when welfare counts only charitable-care patients, but rises to Medicaid-comparable levels when accounting for externalities to the insured.

Keywords: Charity care, uncompensated care, hospital capacity, resource externalities, Medicaid expansion, hospital competition, non-profit hospitals.

JEL Codes: I11, I13, I18, D24, D62, L13, L31

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1 Introduction

Charitable care is a pillar of the American healthcare safety net, providing free or discounted care to those unable to afford it. Charitable care accounts for over one million admissions per year, a volume comparable to that of the Department of Veterans Affairs (Agency for Healthcare Research and Quality, 2025; U.S. Department of Veterans Affairs, 2024), and costs hospitals about thirty billion per year, similar to Workers' Compensation's entire medical spending budget (Centers for Medicare & Medicaid Services, 2026a; Murphy and Welch, 2026). Federal, state, and local governments support charitable care through tax exemptions and indirect compensation programs, thus indirectly insuring the country's 23.5 million uninsured adults (Cohen and Briones, 2026) and millions of underinsured individuals who lack the financial means to pay for care (Collins and Gupta, 2024). However, despite the volume and importance of this care, little is known about its supply or how its provision affects patient welfare.

This article examines the connection between insured and charitable care and how they interact to shape hospitals' care supply decisions. Charitable care is deeply unprofitable and delivered primarily in emergency settings, where regulations constrain hospitals' ability to deny care. In contrast, patient selection in scheduled care is discretionary, allowing hospitals to avoid unprofitable patients. Thus, increased demand for charitable care may discourage hospitals from allocating resources to emergency care and instead incentivize scheduled care for the insured. As the insured also require emergency care, the net effect of charitable care demand on their access is, *a priori*, ambiguous. We refer to the combined impact of one patient's presence on others' access to care, channeled through the use and reallocation of productive resources, as resource externalities.

Using multistate inpatient discharge data from 2010–2019, we estimate the equilibrium consequences of charitable care on hospital resource allocation and patient access to care. We do so by leveraging shocks to charitable care demand to identify its service cost and effect on insured patient access, and then embedding the observed behavior in an equilibrium model of hospital resource allocation and patient demand. Thus, the first step allows us to quantify local effects of charitable care on others, while the second lets us extrapolate to the impact of potential policies. One policy of particular interest directly compensates hospitals for charitable care, allowing us to quantify how underpayment affects hospital resource allocation and access to care. We use the model to evaluate the consequences of insuring otherwise charitable care, and to compute the optimal hospital compensation that trades off fiscal spending, benefits to charitable care patients, and resource externalities to the insured.

Identifying the cost and crowdout effects of charitable care requires resolving two central endogeneity challenges. First, hospitals have discretion over admissions and might admit more patients, including charitable-care patients, when costs are low. Second, charitable-care and insured patients may have correlated preferences for hospitals and needs for care because of shared exposure to hospital marketing or common illness shocks. Resolving these issues requires a shifter

of charitable care demand plausibly exogenous to changes in costs and common preferences.

Our strategy relies on spillover effects from Medicaid access expansions following the passage of the Affordable Care Act (ACA). We show that these expansions led non-profit hospital systems operating in both expansion and non-expansion states to increase their supply of charitable care in non-expansion states. In particular, we estimate that the average non-profit hospital in an expansion state lost 1,543.8 bed-days of charitable care per year following Medicaid expansions. In contrast, their system-siblings in non-expansion states increased care by 1,252.9 bed-days per year. This compensating behavior is consistent with systems' reported charitable care budgets and with their responses to prior charity-care regulation (Almond et al., 2011). We discuss evidence suggesting that incentives to maintain good standing with the IRS, local governments, and donors may drive this behavior. We show that the spillover effects from Medicaid expansion lead charitable care patients in non-expansion states to access better and closer hospitals where supply might previously have been scarce. Moreover, we find that the additional admissions at the hospitals channeling the spillover effects stem from patient reallocation across hospitals, consistent with previous evidence that aggregate demand for charitable care is invariant to moderate supply changes (Garthwaite et al., 2018; Moran, 2025). Thus, for hospitals found near spillover hospitals in non-expansion states, Medicaid expansions in other states produced a plausibly exogenous contraction in the demand for charitable care at their facilities.

These spillovers allow us to build an instrument for charitable care demand and quantify its impact on hospital finances and private care. We estimate that a charitable care admission carries an average service cost of \$13,827, which implies an ex-ante uncompensated care cost of \$1,178 per uninsured individual after accounting for illness risk. This estimate closely matches the (inflation-adjusted) \$1,125 estimate from Garthwaite et al. (2018), based on financial data from 1988–2011. In addition, we find that while the average charitable care patient occupies only 3.8 bed-days per admission, the marginal admission displaces 10.2 bed-days of privately insured care, to an estimated lost revenue of \$36,355. This excess displacement is consistent with hospitals limiting supply because charitable care demand is unprofitable, generating resource externalities for the insured. The revenue impact also shows that hospitals cannot fully shift charitable care costs to insured patients, contributing to the evidence on cost-shifting and cross-subsidization (Cutler, 1998; Frakt, 2011; White, 2013; Clemens and Gottlieb, 2017; Dranove et al., 2017).

To quantify the equilibrium impact of charitable care, we develop a model of hospital competition and inpatient demand. Hospitals in the model allocate limited bed-day resources across service lines (e.g., emergency care) and manage admission requests to each. Bed-days serve as an anchor resource shared across lines, tying production to hospitals' key capital constraint—capacity. Admission decisions succinctly capture how hospitals use queues, rental resources, and flexible staffing. We show that hospitals can use maximum wait-time commitments in a first-come, first-served queue to control their expected admission rate. We use this logic to simplify the service line

problem to choosing an admission rate, subject to expected bed-day use not exceeding allocated resources. The model thus accommodates both observed hospital overflows and their physical and regulatory capacity constraints. In turn, for patients seeking care among hospitals, these decisions determine which hospitals can admit them, endogenizing latent constraints on hospital access.

We estimate the model using patient-level discharge data from Florida, 2010–2019. We show that variation in prices and demand, together with information on hospitals’ total resource constraints, suffices to identify hospitals’ costs. Intuitively, a hospital with multiple service lines allocates resources to equate its marginal profitability across lines, and these elements provide variation in the marginal revenue components necessary to trace differences in costs across lines. This argument yields an estimator of hospital costs that imposes minimal assumptions on prices. Our approach first identifies hospitals’ acceptance rates, following the logic of Agarwal and So-maini (2026), and then combines them with capacity and revenue data to estimate costs.

Our estimates reveal falling hospital markups, skewed resource allocation, and meaningful losses from lack of coordination. As a fraction of cost, we estimate that markups on inpatient care fell from 12.3% in 2013 to 6.3% in 2019, and that while the privately insured pay over twice their cost of service, Medicare rates fall slightly below cost. These structural estimates align with regulatory analyses of hospital finances and prior evidence on private insurance hospital payment (Lopez et al., 2020; Medicare Payment Advisory Commission, 2021). In addition, we estimate that the median hospital accepts 85.9% of all emergency care requests and allocates over half of its available resources to this care. In contrast, it rejects about one in two scheduled charitable-care admission requests and makes only 3% of its bed capacity available to these patients. These admission constraints stem from profitability, as fewer than 5% of hospital-quarters operate at capacity in expectation. We also find that patient allocation leaves gains from trade unrealized because patients cannot trade admission opportunities and prices play no role in allocating care. Notably, we find that the lack of coordination benefits those covered by Medicaid and those seeking charitable care, at the expense of those privately insured and covered by Medicare. Holding supply fixed, we find that the average privately insured Floridian loses the equivalent of one monthly marketplace premium per year due to these factors, while the average uninsured Floridian gains the equivalent of 1.5 monthly marketplace premiums.

To quantify the resource externality of charitable care, we simulate the state’s equilibrium under pseudo-insurance policies for charitable care patients. These policies differ in the compensation they offer hospitals for otherwise charitable care but do not alter other aspects of patients’ access, such as hospitals’ ability to discriminate in providing scheduled care or patients’ networks. At a Medicaid-equivalent compensation level, uninsured patient surplus increases by \$1,235 per patient per quarter from direct access improvements, while the insured gain between \$467 and \$617 on average, depending on their insurance. Because insured patients outnumber uninsured four-to-one in Florida, most of the patient surplus generated by the policy accrues to the insured.

This pattern holds across a wide range of potential compensation levels, with the insured capturing over three-quarters of aggregate surplus gains. However, we also document losses to those seeking insured scheduled care due to the reallocation of resources. In particular, at the Medicaid-equivalent policy, the insured lose access to 5,358 bed-days of scheduled care. We find that competition among hospitals mitigates these losses, as displacing profitable scheduled-care patients from one hospital incentivizes others to increase scheduled-care admissions. In total, the increase in emergency capacity and scheduled charity capacity more than offsets the losses in insured scheduled care, resulting in a 7.46 percentage-point decrease in expected idle capacity.

We find that a policymaker seeking to maximize patient surplus net of fiscal spending would choose to leave the uninsured without coverage only if it ignored the resource externalities to the insured. Intuitively, a policy compensating hospitals for charitable care must also pay for care delivered before any compensation was established. The cost of this inframarginal care is large relative to uninsured surplus gains, since those come not from extensive-margin adjustment but from improved sorting across hospitals, shorter travel times, and better match quality. However, after accounting for positive externalities to the insured, the patient-surplus optimum net of fiscal spending reimburses hospitals at rates 31.4% above Medicaid and well below Medicare, which helps rationalize existing public reimbursement levels. We show that under a wide array of regulatory preferences, including valuing hospital profits, the optimal policy compensates hospitals at or above Medicaid rates, and that in any such outcome most of the surplus would accrue to non-beneficiaries. Competition among hospitals slightly increases the policy's fiscal cost but greatly improves patient surplus, accounting for about one third of the total gains, mitigating the negative consequences of resource reallocations for the insured.

Direct compensation policies, however, are costly and have faced significant political challenges. An alternative of broader political interest imposes minimum requirements on charitable care provided by non-profit hospitals. The challenge, however, is that the benefit of the tax exemption afforded to non-profit hospitals decreases as charitable care delivery increases, because a larger uncompensated pool implies fewer profit dollars are untaxed. This correlation misaligns the benefits and burdens of non-profit status. To address this, academic and policy proposals suggest coupling minimum requirements with a charity care credit market, creating a *floor-and-trade* policy that allows hospitals with high charity burden to finance their care by selling credits to those who provide little (Dranove et al., 2015; Georgia House of Representatives, 2019). Using our model and estimates, we solve for the optimal floor-and-trade policy and find that it attains 88.3% of the surplus gains associated with Medicaid-equivalent pseudo-insurance at zero fiscal cost. We show, however, that the policy must adjust for heterogeneity in demand, resources, and costs across markets and that setting a uniform minimum requirement across hospitals yields minimal surplus gains. Thus, the result highlights a stark tradeoff between targeting and fiscal spending.

Our work bridges the literature on the private and public costs of uninsurance and that on the

value of insurance. Existing research documents the effects of coverage on patients' utilization, health, and financial exposure (Finkelstein et al., 2012; Miller et al., 2021; Dobkin et al., 2018), as well as the incidence of uninsured care on hospitals and other parties (Garthwaite et al., 2018; Finkelstein et al., 2018; Brevoort et al., 2020). Our estimates provide complementary evidence on the costs of uninsurance, and our floor-and-trade policy analysis gives an explicit structure to the role of non-profit hospitals as insurers, as argued by Garthwaite et al. (2018). Our evidence of private patient crowdouts and our theory of resource allocations rationalize prior mixed findings regarding the effect of uncompensated care on insured patient access and contribute to a broader literature on the supply effects of insurance and spillovers across patient groups (Finkelstein, 2007; He et al., 2015; McInerney et al., 2017; Glied and Hong, 2018; Carey et al., 2020). In particular, our theory shows that a higher uninsured burden might increase or decrease insured patient admissions, depending on the relative uninsured load and the comparable profitability of scheduled care.

Our welfare analysis adds the resource-allocation channel to analyses of insurance value and incidence (Finkelstein et al., 2019a,b; Hendren, 2021; Hendren and Sprung-Keyser, 2020). As in Finkelstein et al. (2019b), we find that beneficiaries' willingness to pay for coverage falls below the gross cost of providing it. Incorporating the benefits to non-recipients, however, changes the sign of the estimated welfare effect of expanding coverage. Our analysis also complements concurrent research by Ghili et al. (2026) on the impact of uncompensated care on insurance take-up. While we focus on the resource externalities of charitable care, they focus on its effect on demand for formal insurance. Both articles document externalities that provide an additional rationale for collective intervention in insurance markets with implicit coverage through uncompensated care (Coate, 1995; Mahoney, 2015), and connect to work showing how insurance externalities alter optimal policy (Chandra et al., 2010; Cabral and Mahoney, 2019). Our analysis of how resource externalities affect optimal pseudo-insurance contracts also contributes to the literature on insurance design and the welfare consequences of insurance-market interventions (Einav et al., 2010; Hackmann et al., 2015; Marone and Sabety, 2022; Ghili et al., 2024; Tebaldi, 2025; Vatter, 2025).

This article also contributes to the literature on hospital competition and healthcare production. Our analysis of capacity constraints and patient allocation builds on classical work on reserve capacity and uncertain demand by Joskow (1980) and Gaynor and Anderson (1995). Our model complements the literature on hospital pricing behavior (Gaynor and Vogt, 2003; Gowrisankaran et al., 2015; Ho and Pakes, 2014; Ho and Lee, 2017; Handel and Ho, 2021) by having hospitals optimize over the supply of care rather than its price. The analysis accounts for capacity constraints and overflow costs, which are necessary to explain crowdouts and observed allocation. However, we abstract from potential effects of congestion on treatment quality (Hoe, 2022; Gowrisankaran et al., 2026). Our focus is on how resource scarcity affects costs and allocation across patients and competing hospitals. Although waiting times motivate our interpretation of rationing, we leave them implicit because our data do not directly measure them. Our work is therefore complementary to that on how wait times affect choice, rationing, and patient outcomes (Sivey, 2012;

Gruber et al., 2023; Russo, 2024). Hospitals’ discretion over admissions for scheduled charitable care also connects our analysis to the provider-selection literature (Newhouse, 1996; Ellis, 1998). Particularly close is Gandhi (2026), who studies selective admissions in nursing homes and the opportunity cost of committing beds to patients with long expected stays. His analysis emphasizes the possibility of displacing more profitable future admissions; ours emphasizes the opportunity cost of allocating shared resources across contemporaneous hospital services. More broadly, our work adds to growing evidence that firms facing constraints on consumer selection and declining margins on one output reallocate resources to others, as documented in housing (Diamond et al., 2019), education (Bound et al., 2020), and banking (Chakraborty et al., 2018; Bidder et al., 2021).

Finally, our work contributes to the literature on non-profit hospital behavior and charity-care provision. A longstanding debate concerns whether hospital ownership changes providers’ objectives and behavior (Newhouse, 1970; Weisbrod, 1988; Frank and Salkever, 1991; Sloan et al., 2001; Duggan, 2002; Horwitz and Nichols, 2009; Liu, 2022). Evidence that non-profit hospitals respond to financial incentives much like for-profit hospitals has motivated a variety of models of firm behavior, some focusing exclusively on admission profits and others incorporating preferences for quality, volume, and social surplus (Gaynor and Vogt, 2003; Gowrisankaran et al., 2015; Ho and Lee, 2017). Our estimates suggest non-profit hospitals place positive weight on charity-care delivery consistent with models of (impure) altruism (Frank and Salkever, 1991). Our evidence, however, cannot distinguish altruism from an implicit quid pro quo with regulators (Fournier and Campbell, 1997), preservation of tax-exempt status, donor appeal, or political goodwill.

The remainder of the article is organized as follows. Section 2 presents a conceptual framework that highlights the economics of resource externalities. Section 3 provides further details on the setting and data used in this article. Section 4 presents our main evidence from Medicaid spillovers. Section 5 presents our empirical model, and Section 6 discusses its identification and estimation. Section 7 presents our key structural estimates, including markups, resource allocation, and allocative efficiency. Section 8 presents the counterfactual policy analysis quantifying the resource externalities of charitable care. Finally, Section 9 concludes.

2 Conceptual Framework

We begin by illustrating the key economic forces behind our analysis, explaining how the insured and uninsured interact in the allocation and availability of hospital resources. For simplicity, we abstract from competition and heterogeneity and consider a profit-maximizing monopolist hospital facing demand from a unit mass of homogeneous patients. A fraction λ of the population requires emergency care, and an independent fraction φ lacks insurance. Each admitted patient occupies one bed, costs the hospital c to serve, and, if insured, yields a revenue of p . Uninsured patients, by contrast, leave care uncompensated. The hospital has B beds available for service and must allocate them between emergency and scheduled care. Emergency care is non-discriminatory,

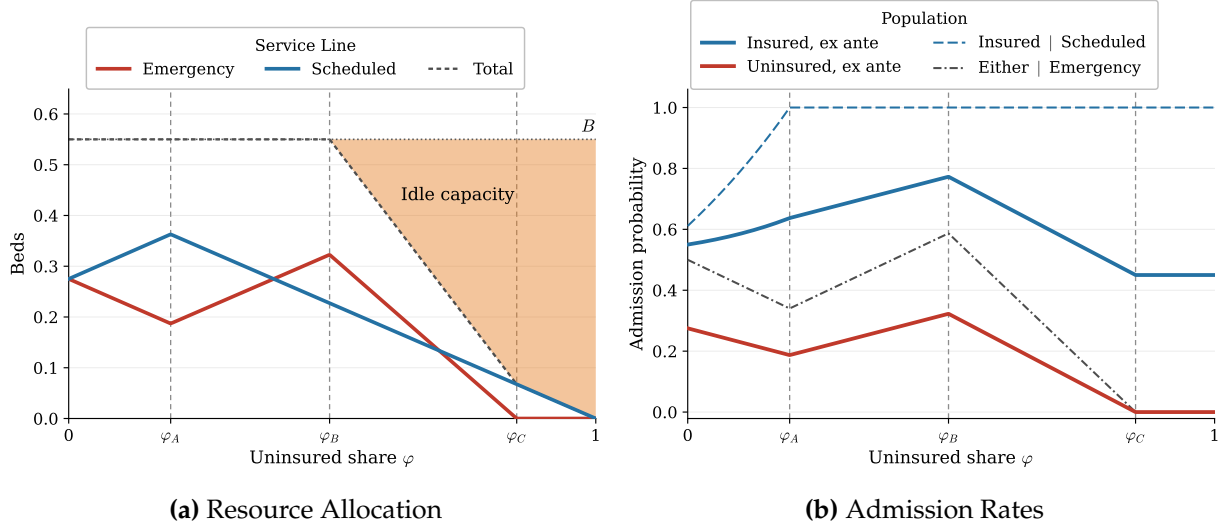


Figure 1: Conceptual Framework – Resource Allocation Under Uninsurance Burden

Notes: These figures show how a monopolist hospital allocates resources between its emergency and scheduled care service lines as a share φ of its population is uninsured. The plots are based on the solution of Equation (1) with $B = 0.55$, $\lambda = 0.55$, $p = 1$, $c = 0.15$, and $c^V = 0.55$. Capacity is set below total potential demand to display all four regimes.

and the hospital must accept patients as long as it has capacity. In contrast, scheduled care is discriminatory and thus excludes the uninsured from care.¹

In this simple environment, the hospital's problem reduces to deciding the expected number of patients to admit to each line subject to the total available population and its capacity constraint:

$$\max_{\substack{0 \leq q_E \leq \lambda \\ 0 \leq q_S \leq (1-\lambda)(1-\varphi) \\ q_E + q_S \leq B}} \sum_{j \in \{E, S\}} (\pi_j q_j - c^V q_j^2) \quad (1)$$

where q_j is the number admitted to line j , c^V is the additional congestion cost, and π_j is the line's expected profit per patient: $\pi_E = (1 - \varphi)p - c$ and $\pi_S = p - c$. This formulation allows analytic solutions and visual exploration, as shown in Figure 1. Given the hospital's constraints, its behavior is segmented into four regimes, each with distinct comparative statics linking uninsured share and optimal admission behavior.

In the first regime, uninsurance is low ($\varphi \leq \varphi_A$), and only the hospital's total capacity constraint binds. As the rate of uninsurance increases, expected profits from emergency admissions decline relative to those from scheduled care, leading the hospital to reallocate resources as seen in Figure 1a. This reduces emergency admissions by all patients and increases those from insured scheduled care, as seen in Figure 1b. At φ_A , the hospital enters its second regime, where its de-

¹As in our application, the hospital has no other means to attract or dissuade patients. However, extending the mechanisms described here to a scenario with price or quality adjustments is straightforward as long as these do not allow the hospital to screen out the uninsured.

mand for scheduled care has been exhausted, but its emergency demand is sufficiently populated by the insured that it is still profitable to assign it resources. From φ_A to φ_B , emergency admissions increase one-for-one as demand for scheduled insured care decreases. At the same time, an increasing share $\varphi(1 - \lambda)$ demands scheduled care without insurance and is left unserved. As the uninsured share increases, the marginal profitability of emergency admissions declines, reaching zero at φ_B . At this point, in the third regime, the hospital's capacity constraint no longer binds as it gradually shuts down emergency services. In the final regime, at φ_C , the emergency department closes, and the hospital becomes a small clinic for the few insured left.

This illustration highlights the resource externalities of uninsurance. At low ($\leq \varphi_A$) and high ($\geq \varphi_B$) levels, increasing uninsurance reduces resource allocation to emergency care, reducing access for all patients. However, at low levels of uninsurance, insured access to scheduled care increases with uninsurance, implying that the externalities of uninsurance depend on the insured's relative need for emergency care and on their value from care at the monopolist hospital relative to the outside option. In contrast, at high levels of uninsurance, insured welfare systematically declines with uninsured rates. The figure also reveals a region, (φ_A, φ_B) , where a higher uninsured rate leads to higher emergency care admission rates, creating positive externalities for the insured. Thus, the sign of resource externalities depends on which of the four regimes dominates market behavior, which in turn depends on the empirical primitives of each setting.

Costs aside, access to insurance benefits those who would otherwise go uninsured and potentially those who are insured regardless. As illustrated here, patients inframarginal to changes in insurance coverage are affected by how these policies affect access. This illustration highlights how resource allocation changes with insurance access, who might benefit from broader access, and who might prefer to keep hospital care exclusive. The forces governing the impact of insurance coverage depend on hospitals' costs, revenues, and patient preferences and needs. As in this analysis, the rest of this work abstracts from any additional value uninsured patients might derive from risk protection under counterfactual insurance and focuses on how such insurance affects payments and access, framing the question as regulating payments and access to otherwise charitable care.

3 Setting and Data

This section describes the regulatory environment of charitable care delivery and non-profit hospital status, as well as the data we use for our analysis. We focus on Florida for state-specific examples, as it is the setting for our structural analysis. For a review of the history of access to insurance in the US, see Gruber (2008), and for a summary of changes leading to and following the expansion of Medicaid access under the ACA, see Gruber and Sommers (2019).

3.1 Setting

Charitable care refers to services provided for free or at significant discounts to patients with limited ability to pay.² The geography of this care is tied to insurance access, with states that did not expand Medicaid after the ACA having higher uninsurance and charity care rates (Levinson et al., 2025). For example, Florida did not expand Medicaid access while Massachusetts did. Florida has an uninsured rate of 15.5% among working-age adults and 267,000 non-disabled, non-elderly adults ineligible for the state’s Medicaid program and with incomes too low to qualify for subsidies on the ACA insurance exchanges (KFF, 2024). Massachusetts, in contrast, has an uninsurance rate of 3.7%, and Medicaid is broadly accessible for legal residents who do not qualify for ACA subsidies.³ Florida’s hospitals also spend 2.1 times more on uncompensated care per capita than those in Massachusetts (Centers for Medicare & Medicaid Services, 2026a).

Charitable care is primarily delivered in emergency settings, where federal and state regulations constrain hospitals from discriminating among patients. In particular, the federal Emergency Medical Treatment and Labor Act requires Medicare-participating hospitals with an emergency department to provide screening and stabilization to any patient with emergency medical conditions, regardless of insurance or ability to pay (United States Congress, 2023). It also prohibits delaying screening or care because of these factors and requires hospitals to accept transfers when capacity permits. For tax-exempt non-profit hospitals, section 501(r)(4)(B) of the Internal Revenue Code requires them to establish an emergency-care policy that provides care without discrimination (Internal Revenue Service, 2014). In addition, many states impose stricter requirements. For example, both Florida and California require emergency care admissions and transfers to be accepted, as long as capacity permits, regardless of insurance or economic status, and forbid requiring payment guarantees as conditions for care (Florida Legislature, 2023; California Legislature, 2019).

Beyond hospital costs, charitable care carries a high public cost through programs that incentivize and compensate hospitals for this care. One such program is the federal 501(c)(3) non-profit status, under which hospitals are expected to provide community benefits, including the provision of charitable care. In return, these hospitals obtain a variety of benefits, including exemption from federal income taxes and, in most cases, from state income and sales tax as well as local sales and property taxes (Plummer et al., 2024; Godwin et al., 2023). In 2020, non-profit hospital tax benefits were estimated at \$28.1 billion, exceeding that year’s reported charitable care spending of \$16 bil-

²Charitable care differs from bad debt, which arises from patients the hospital attempted to collect from but could not. Charitable care, instead, is usually determined before collection and sometimes before the patient is admitted. Uncompensated care is the sum of these terms.

³As of 2026, Florida’s income limit for Medicaid qualification for parents in a family of three was 26% of the federal poverty level, while other non-disabled adults had no income-based eligibility pathway. In Massachusetts, the effective income limit is 138% of the federal poverty line. Citizenship, immigration, residency, and other eligibility requirements also apply, so the absence of a statutory coverage gap does not imply universal take-up.

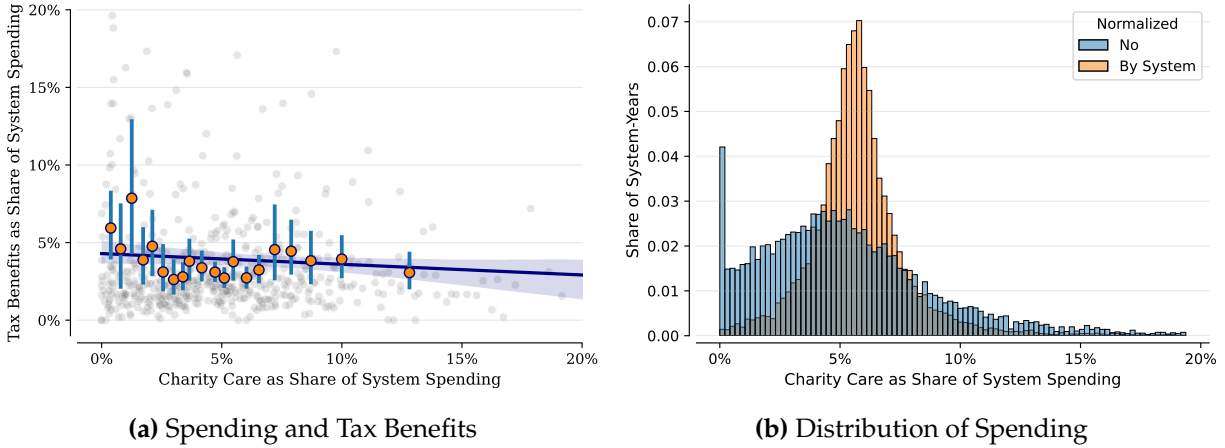


Figure 2: Charitable Care Spending at Non-Profit Hospitals

Notes: These figures display the distribution of charitable care spending for non-profit hospitals, at the system-year level. Figure 2a shows tax benefits, computed from available public tax data, and charitable care spending, as reported in the systems' tax returns. Both are measured in 2019. The gray dots correspond to the underlying data. In contrast, the orange dots correspond to mean estimates within ventiles of the x-axis, and the line represents the overall sample regression line. Figure 2b shows the distribution of charitable care spending of systems observed for at least three years in our sample. The unnormalized bars (blue) show the raw distribution, while the normalized bars (orange) are the residuals after controlling for a system fixed effect, recentered to the sample mean. Both figures restrict to systems with at least \$100 in system-level spending in a year.

lion (Godwin et al., 2023).⁴ Non-profit hospitals are required to form clear and public plans for the provision of free and discounted care, have clear guidelines for who qualifies for this care, conduct a Community Health Needs Assessment at least once every three years to determine the community's needs for care, and follow strict procedures for evaluating whether patients qualify for assistance before any attempt to pursue extraordinary collection actions (Internal Revenue Service, 2026a,c). Non-compliant hospitals face public disclosure of failures, a potential excise tax, income taxation, and, in very serious cases, revocation of federal exempt status (Internal Revenue Service, 2026b). Enforcement, however, is gradual and correction-oriented, with over three-quarters of financial assistance policy compliance checks resulting in corrective action (Internal Revenue Service, 2015; Treasury Inspector General for Tax Administration, 2025).⁵

However, the costs and benefits of non-profit status are often misaligned. As shown in Figure 2a, the correlation between systems' tax-exemption benefit and their spending on charitable care is negative. Intuitively, serving the uninsured reduces profits and thus the system's taxable dollars, so those providing the most stand to benefit the least from the tax exemption. Notably, the

⁴We describe tax benefits and public spending using the most recent available data when possible. However, because of gaps and inconsistent reporting, some numbers below span different years, which we make explicit.

⁵Revocations are rare but do occur. For example, in 2017, the IRS revoked a hospital's 501(c)(3) recognition after finding serious failures in meeting program requirements (Internal Revenue Service, 2017). State enforcement also occurs, as in the case of Provena Covenant Medical Center v. Department of Revenue, in which the hospital lost its property-tax exemption, in part, because its charity-care program was inadequately publicized and patients were billed before being considered for assistance (Supreme Court of Illinois, 2010).

IRS does not dictate a minimum level of charitable care or community benefits, and government analysis documents that some systems had no charitable care or community benefit spending (U.S. Government Accountability Office, 2023). Yet these gaps cannot be attributed solely to hospital decisions. Hospitals' status depends on community benefit provision guidelines, but their communities are not required to demand these benefits. Thus, some of those benefiting the most from tax exemption in Figure 2a are simply located in heavily insured areas.

Nevertheless, non-profit systems budget for charitable care and operate around seemingly persistent spending targets. In federal tax filings from 2011 to 2019, 93.3% of systems reported budgeting for charitable care, 51% reported expenses exceeding their planned budget, but only 0.23% reported denying care to eligible patients because of it. System fixed effects explain 59.2% of the variance in spending, as shown in Figure 2b, suggesting some persistence in planning. This budgeting and target-setting behavior is also consistent with hospitals' responses to prior regulation on charitable care (Almond et al., 2011). We record this persistence as a feature of the data and take no stand on its source. It informs the analysis conducted in Section 4, and we provide further evidence supporting it at the end of that section.

Beyond tax exemption, several other state and federal programs compensate hospitals for charitable or uncompensated care. In 2024, Medicare spent \$9.28 billion on Disproportionate Share Hospital (DSH) payments, with \$5.94 billion compensating hospitals for uncompensated care (Medicare Payment Advisory Commission, 2026). Similarly, Medicaid spent \$12.6 billion on its DSH program, including federal and state funds, to compensate hospitals for Medicaid shortfalls and uninsured patient costs (Medicaid and CHIP Payment and Access Commission, 2026). However, these programs rarely affect hospitals' decisions on the margin because of lags in their effects, uncertainty about budget distribution, and other factors. For example, an additional dollar of current uncompensated care affects Medicare payments only through the hospital's future share of a fixed national pool, after a reporting and audit lag, and the amount depends on uncompensated-care reporting by every other eligible hospital. Similarly, Medicaid DSH payments are assigned to the state to redistribute, resulting in allocations that do not reflect the burden of charitable care. For example, non-profit hospitals in Florida accounted for 70.9% of all uncompensated care in 2014 but obtained only 18.7% of that year's payments, with the vast majority going to public hospitals (U.S. Government Accountability Office, 2019). Other programs offer compensation to hospitals, but to our knowledge, none provided meaningful marginal compensation during our period of analysis, 2010 to 2019. However, put together, public spending on charitable care is considerable and has often exceeded reported costs.

3.2 Data

The primary dataset for our analysis is a discharge-level panel of inpatient admissions from the State Inpatient Databases (SID) of the Healthcare Cost and Utilization Project (HCUP) for 2010–

2019. The dataset covers 18 states and includes discharges from all patient types, including those uninsured. We follow standard practice by associating charitable care with those patients who, at discharge, had a fully forgiven bill or were registered as self-pay, which aligns with hospital coding practices (Cram et al., 2010; Popescu et al., 2019; Barrett et al., 2021). We require records to have non-missing hospital identifiers, primary payer type, diagnosis-related group (DRG) code, length of stay, and admission year and quarter. We collect patients’ five-digit ZIP codes, admission types, and emergency department usage, and retain only data from states that report both before and after 2014, the first year of ACA Medicaid expansions.

We complement the discharge data with datasets on hospital characteristics, finances, and merger activity. We obtain hospital beds, ownership, system affiliation, and additional attributes from the American Hospital Association (AHA) Annual Survey. We link the data to the hospital merger panel developed by Brot et al. (2024), which documents M&A activity and allows us to control for confounding changes in hospital ownership. We obtain hospitals’ reported costs and revenue from the Healthcare Cost Report Information System (HCRIS). Finally, we complement the data with information from Schedule H of IRS Form 990, which reports the gross and net cost of each community-benefit category, including financial assistance programs or charitable care.

Because the SID reports only provider charges, we supplement the payment data. First, for Medicare and Medicaid admissions, we use CMS’s payment formulas, as specified in the Final Rule and Correction Notice files for each fiscal year, and apply Florida’s specific Medicaid rules. These payments include those from compensating programs such as DSH. For private insurers, we use the Florida Health Price Finder files, which report price percentiles for common procedures at each hospital as paid by private insurance enrollees. We follow longstanding evidence that hospitals negotiate private contracts as a multiple of Medicare rates or a percentage discount off charges (Cooper et al., 2019), determine the best-fitting hypothesis for the published private rates, and use it to extend prices to unreported procedures. We use these payments only for our structural analysis. Finally, we rely on non-profit system tax returns to obtain the average direct payment hospitals receive from patients for charitable care, which averages to 5.68% of Medicaid payments or about \$308 per charitable care patient. We use these rates to compute expected direct compensation from charitable care patients in the structural analysis. Unless otherwise stated, monetary values from our analysis are in CPI-adjusted 2023 dollars.

4 Expansion Spillovers and Charitable Care Costs

This section documents cross-state spillovers from Medicaid expansion and uses them to identify charitable care costs and insured patient crowdouts. Our strategy leverages a channel created by hospitals in non-expansion states that are part of systems also found in expansion states. Figure 3 shows our treatment taxonomy, which we use throughout to distinguish hospitals. We refer to hospitals in expansion states as *directly treated* and those in non-expansion states with system

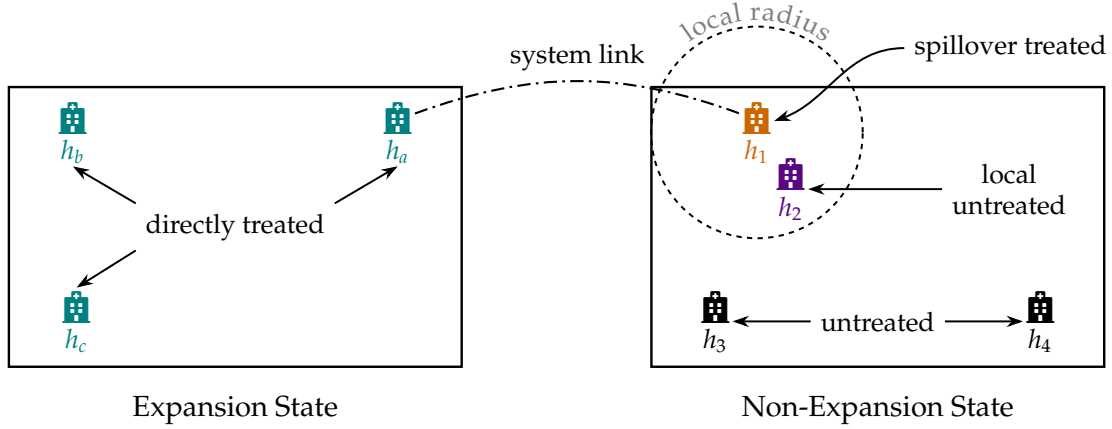


Figure 3: Treatment Taxonomy

Notes: Hospitals in expansion states are referred to as *directly treated*. Hospitals in non-expansion states belonging to systems with a directly treated hospital are *spillover treated*. Hospitals in non-expansion states that are not part of systems with members in expansion states and are located within the specified local radius of a spillover-treated hospital are referred to as *local untreated*. We refer to other hospitals in non-expansion states simply as *untreated*.

siblings in expansion states as *spillover treated*. We refer to hospitals in non-expansion states located near spillover-treated hospitals and that are not part of systems with members in expansion states as *local untreated*. All other hospitals in non-expansion states are simply *untreated*.

We focus on local untreated hospitals. We hypothesize that system-level objectives at non-profit hospitals will lead spillover-treated hospitals to expand charitable care supply after demand for these services falls in their linked, directly treated hospitals. If aggregate demand for charitable care in non-expansion states does not increase substantially, the local untreated hospitals will experience the supply increase at spillover-treated hospitals as a demand contraction. This demand shift will identify their cost curve and the effect of charitable care admissions on their insured patients.

We build toward the main evidence in steps. First, we document spillover effects, focusing on spillover-treated hospitals. Second, we show that aggregate demand did not meaningfully change and that demand was reallocated, which requires examining the local markets and states in which spillover-treated hospitals are located. Finally, we build the instrument for charitable care demand at local untreated hospitals and use it to measure the effects there. We conclude this section with an analysis of motives and mechanisms that might underlie our variation.

4.1 Data Structure and Exclusions

We rely on a subset of our data for this analysis, removing entries that might experience confounding variation and restructuring the panel for our empirical strategy. First, to avoid changes due to patient reallocation, we consider only hospital admissions by patients living in the same state as the hospital. Second, to avoid changes due to imminent closures and to maintain a comparable and relevant set of hospitals, we focus on general acute care hospitals with at least 50 beds that

do not exit the market during our analysis. Third, to limit acquisition-related confounding, we exclude direct- and spillover-treated hospitals acquired within a seven-year window centered on the relevant expansion event, and we exclude untreated hospitals that are ever part of treated systems. This no-acquisition restriction is particularly strict for for-profit hospitals in non-expansion states, dropping almost all of them from our sample. To keep our analysis consistent, we exclude all for-profit hospitals from the regressions presented in this section.

To study the spillover effects, we structure the data into stacked and balanced event panels. This allows us to form control groups for each directly or spillover-treated hospital by randomly matching it to hospitals from the untreated group.⁶ We require control hospitals to be continuously observed and not undergo acquisitions within the same windows as for treatment units. Control units capture trends in demand for, and supply of, charitable care services. For spillover-treated hospitals, we rely on others within the same state to account for national and local charitable demand changes and incentives to provide it. For directly treated hospitals, we match to other system hospitals in non-expansion states to account for national trends. We match treated hospitals to a random sample of potential controls based on bed-size similarity and focus on their first treatment period: the first time their system experiences a Medicaid expansion.⁷

4.2 Spillover Effects

We begin by examining if and how Medicaid expansions changed the demand for and supply of charitable care. Using our stacked panel of events, we conduct regressions of the form:

$$y_{eht} = \alpha \mathbb{1}\{t \geq 0\} \mathbb{1}\{h = h_e^*\} + \gamma_{eh} + \delta_{et} + \varepsilon_{eht} \quad (2)$$

where h indexes hospitals, t indexes years relative to treatment (i.e., the expansion event), and e indexes treatment events. The outcome for hospital h , t years after treatment, is denoted y_{eht} , and γ_{eh} and δ_{et} are event-hospital and event-time fixed effects. Each event has a unique treated hospital, identified by h_e^* , and a balanced set of control units. We also use the standard event-study form, estimating a separate α for each period relative to $t = -1$.

⁶Our sample and regressions are designed to contend with known challenges in two-way fixed-effects analysis (Callaway and Sant’Anna, 2021; Baker et al., 2022). Following Wing et al. (2024), we weight our regression so each treated unit gets a weight of 1, and each control unit matched to an event e gets a weight of $1/N_e$, where N_e is the number of matched controls. To account for the stacked structure, we estimate all regressions using two-way clustered standard errors, clustering at the event and hospital levels.

⁷For spillover-treated hospitals, we define system size categories by the median total system bed count within each state in 2014. To avoid mechanical inflation in estimated effects from demand substitution, we exclude control hospitals located within 30 miles of treated hospitals. We also require hospitals to share the same Medicare accreditation status as the target hospitals. For direct treatment, we match size using 50-tiles of 2014 bed size. We use this stricter match for direct treatment since the pool of potential controls is substantially larger than for spillover treatments, allowing us to match hospitals that are more similar to each other. We exclude hospitals in states that expanded Medicaid access by 2020 from the potential set of spillover-treated, local-untreated, or untreated hospitals. We exclude spillover hospitals in states that eventually expand Medicaid.

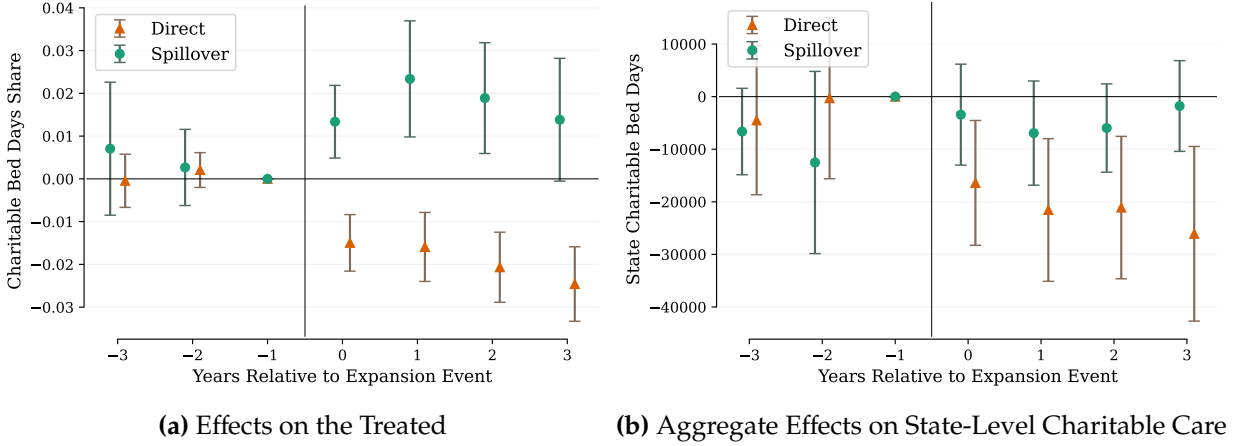


Figure 4: Direct and Spillover Effects From Medicaid Expansion on Charitable Care

Notes: Figure 4a shows the estimated effect of Medicaid expansion on the charitable-care share of bed-days at hospitals in expansion states (triangles) and at hospitals in non-expansion states linked to expansion states through system membership (circles). These estimates match the event-study variant of the stacked event-study panel regression (2), with standard errors clustered at the event and hospital levels. Figure 4b shows the estimated effect of expansion on overall charitable care volume in expansion states (triangles) and in non-expansion states subject to spillover effects (circles). These estimates match the event-study variant of the state-year-level regression (3), with heteroskedasticity-robust standard errors. In both figures, we include only patients residing in the same state as the affected hospitals in the demand counts. Error bars represent 95% confidence intervals.

Figure 4a shows the estimated effect of Medicaid expansion on charitable care in directly and spillover-treated hospitals. To make changes comparable across hospitals, we first measure charitable care as the share of total bed-days served at each hospital dedicated to charity patients.⁸ Directly treated hospitals experience a sharp 1.9 percentage-point decline in the relative volume of charitable care following Medicaid expansion, equivalent to a 43.2% decrease relative to their pre-expansion baseline. In contrast, spillover-treated hospitals experience a 1.4 percentage-point increase in the relative volume of charitable care, amounting to a 25.5% gain relative to their pre-expansion baseline. The first two columns of Panel A in Table 1 show the supporting estimates, and columns 3 and 4 show the effect measured in bed-day counts. Both show clear opposing changes between the two sets of treated hospitals, a pattern that persists when measuring admission counts or total DRG weights of admissions (see Appendix Table A.1). Columns 5 and 6 also show that directly treated hospitals experience a more-than-offsetting increase in demand from Medicaid patients—as expected given expanded access to public low-income insurance—while spillover-treated hospitals show no significant change in Medicaid bed-days.

In Appendix B.1 we provide additional evidence on how these spillovers affected charitable care patients. First, relative to privately insured patients, charitable care patients in counties with at least one spillover-treated hospital are admitted to hospitals with better performance on a variety of mortality outcomes after treatment. Second, we show that the average distance charitable care

⁸We weight admissions volume by the number of bed-days used, as we are interested in understanding how one patient crowds out resources for other patients. Because patients' length of stay varies with medical needs, which correlate with insurance status, comparing admission counts may misrepresent how constrained hospital resources are.

Table 1: Medicaid Expansion Effects on Charitable Care

Panel A: Expansion Treatment Effects on Care Delivered (bed-days)								
	Charitable Share		Charitable		Medicaid		State Charitable	
	Direct (1)	Spillover (2)	Direct (3)	Spillover (4)	Direct (5)	Spillover (6)	Direct (7)	Spillover (8)
Expansion Effect (α)	-0.019*** (0.003)	0.014*** (0.004)	-1543.808*** (316.991)	1252.872*** (467.916)	3801.132*** (836.559)	-158.309 (710.476)	-19681.283*** (3736.339)	1859.326 (3438.394)
Mean Dep.	0.044	0.055	2748.671	3938.259	14008.759	13709.186	37229.152	77643.906
N	2081	3065	2081	3065	2081	3065	231	77
R ²	0.914	0.908	0.944	0.962	0.994	0.995	0.866	0.993

Panel B: Demand Reallocation in Non-Expansion States (bed-days)								
	Charitable		Gov.		Private		Total	
	Spillover (1)	Local (2)	Spillover (3)	Local (4)	Spillover (5)	Local (6)	Spillover (7)	Local (8)
Expansion Effect (α)	1252.872*** (467.916)	-1009.057** (424.829)	345.021 (2316.662)	1173.584 (1604.996)	-1473.660 (1453.212)	1681.869* (909.450)	486.549 (3654.607)	2807.092 (1963.974)
Mean Dep.	3938.259	8442.128	50885.332	63565.617	15247.444	23601.596	72637.555	98385.344
N	3065	1156	3065	1156	3065	1156	3065	1156
R ²	0.962	0.944	0.996	0.988	0.989	0.974	0.996	0.992

Notes: "Local" stands for local untreated. All measures include only patients residing in the hospital's state. Care delivery is measured in bed-days based on admitted patients' length of stay. "Charitable Share" is the total length of stay of charitable care patients relative to the total length of stay of all admitted patients at the hospital. "State Charitable" corresponds to the total length of stay of all charitable care patients in the state. "Gov." stands for Medicare and Medicaid admissions. Columns 1–6 of Panel A and all spillover columns in Panel B are estimates from Equation (2), with two-way clustered standard errors at the event and hospital levels. Columns 7 and 8 of Panel A are estimates from Equation (3), with heteroskedasticity-robust standard errors. See Appendix Figure A.1 for the supplemental event-study graphs for Panel A. Panel B local estimates are based on Equation (4). "Mean Dep." for Direct and Spillover columns is the treated-unit mean in the year before expansion treatment. For Local columns, it is the unweighted mean among exposed hospitals in the year before their first exposure. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

patients travel to spillover-treated hospitals decreases after treatment. Third, we show that the added charitable care admissions at spillover-treated hospitals are not for hard-to-redirect medical conditions, such as circulatory problems or pregnancies. Instead, the increases are driven by additional care for patients with HIV, cancer, and immune diseases, among others. Together, these suggest that expansion spillovers allowed some charitable care patients to access hospitals that were previously inaccessible to them and preferred over their alternatives. As we show next, aggregate demand for charitable care in non-expansion states did not change with spillover treatment, suggesting that these improvements in patient access are driven by increased supply.

4.3 Demand Reallocation

Consistent with uninsured individuals accessing Medicaid coverage in expansion states, aggregate charitable care admissions declined sharply in expansion states, but did not change in non-

expansion states. We examine this using a regression at the state-year level:

$$y_{st} = \alpha \mathbb{1}\{t \geq t(s)\} + \zeta_s + \varepsilon_{st} \quad (3)$$

where y_{st} is a state-year outcome, $t(s)$ is the state's first year of direct or spillover treatment, and ζ_s is a state fixed effect. Figure 4b shows the estimated effects per year, while columns 7 and 8 of Panel A in Table 1 show the supporting aggregate estimates. Appendix Table A.1 shows a similar pattern for state-level charitable care admissions and spending, with decreases in expansion states and no statistically significant changes in non-expansion states.

The observation that aggregate demand for charitable care in non-expansion states did not increase implies that the added care at spillover-treated hospitals must stem from a reallocation of charitable care patients across hospitals within the state. This largely rules out spillover-treated hospitals increasing charitable care by offering it to patients who would otherwise have relied on insurance or not sought care altogether. Any such change would necessarily increase aggregate charitable care in spillover states.

To further explore this reallocation hypothesis, we examine whether local untreated hospitals experienced meaningful changes in patient composition after their neighbors' spillover treatment. We focus on untreated hospitals located within 15 miles of a spillover-treated hospital. To align with our analysis and mitigate merger-related confounding, we adopt the same data requirements imposed on treated hospitals in our stacked event panel.⁹ In addition, to avoid mechanical effects due to patient overflow and local transfer agreements, we exclude hospitals found within 5 miles of spillover-treated hospitals.¹⁰ Using this sample, we estimate the model:

$$y_{ht} = \alpha T_{ht} + \gamma_h + \delta_{s(h)t} + \varepsilon_{ht} \quad (4)$$

where y_{ht} is the outcome, T_{ht} indicates whether the hospital was exposed locally to a spillover effect at a period prior to or equal to t , γ_h is a hospital fixed effect, and $\delta_{s(h)t}$ is a state-year fixed effect. This last fixed effect implies that α captures how local untreated hospitals' demand changed relative to other untreated hospitals in the same state, following the taxonomy in Figure 3.

Panel B of Table 1 reports the estimates and compares them with outcomes at spillover-treated hospitals. The first two columns indicate that local untreated hospitals experience changes opposite to those measured at spillover-treated hospitals. While the average spillover-treated hospital increases charitable care by 1,253 bed-days, the average local untreated hospital decreases it by

⁹Specifically, we require hospitals to be general acute care hospitals, have at least 50 beds, not undergo an acquisition within a seven-year window centered on treatment, be not-for-profit, and not exit by the end of our sample.

¹⁰See Appendix Table A.15 for the same analysis without the 5-mile filter and with the additional requirement of hospitals being system hospitals, to increase comparability with spillover-treated hospitals. Appendix B.2 discusses these results and provides additional evidence by emergency status and insurance source.

Table 2: Charitable Care Effects at Local Untreated Hospitals

	Panel A: Effects on Insured Care (IV) (bed-days)							
	Overall		Emergency Care			Scheduled Care		
	Gov.	Private	Total	Gov.	Private	Total	Gov.	Private
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Charitable Care Admissions (β)	-2.198 (2.120)	-8.049*** (2.323)	-5.047*** (1.631)	-0.475 (1.194)	-4.572*** (1.112)	-5.200** (2.493)	-1.723 (1.509)	-3.477*** (1.282)
Mean Dep.	63565.617	23601.596	46258.277	36798.787	9459.489	40908.936	26766.830	14142.106
N	1156	1156	1156	1156	1156	1156	1156	1156
R ²	0.986	0.953	0.963	0.967	0.894	0.973	0.967	0.964

	Panel B: Effects on Cost and Revenue (IV) (thousands)							
	Profit	Revenue			Cost			
	Total	Total	Medicare	Private (proxy)	Total	Labor	Uncompensated Care	Charity Care
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Charitable Care Admissions (β)	-29.160*** (7.577)	-36.335*** (8.186)	-2.770** (1.171)	-49.052*** (9.837)	-7.175** (3.466)	-27.861 (17.249)	15.793** (6.895)	13.827** (6.859)
Mean Dep. (millions)	42.806	263.981	78.850	150.250	221.176	204.553	54.509	36.461
N	977	977	537	537	977	977	977	977
R ²	0.862	0.983	0.988	0.949	0.991	0.984	0.662	0.504

Notes: Estimates from Equation (5), with charitable care admissions instrumented by local exposure to spillover effects. In Panel A, “Gov.” is Medicare plus Medicaid, “Total” is government plus private, and scheduled is the complement to emergency care. In Panel B, profit, revenue, and total cost are for inpatient care only; the other cost measures include outpatient care. Medicare revenue is inpatient reimbursement from HCRIS Worksheet E, Part A, line 71. We proxy revenue from private insurers using financial data, as described in Appendix B.3. We cap each payer-specific revenue measure and labor cost at the 97.5th percentile. “Mean Dep.” is the unweighted mean for exposed hospitals observed the year before their first local exposure. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

about 1,009 bed-days, or 12.0% relative to the year before first exposure. Similarly, while spillover-treated hospitals see a noisy decrease in private admissions, local untreated hospitals see an increase of approximately 7.1% relative to baseline. The precision of this potential reallocation of insured patients is limited, likely because of heterogeneity in the intensity of the local spillover effects and local insurance composition. Together, this evidence motivates using spillovers as an instrument for charitable care demand at local untreated hospitals, but with a variable that captures spillover intensity, as we do next. The effect on charitable care reallocation presented here serves as the first stage of said instrument.

4.4 Crowdouts and Private Costs of Charitable Care

We now turn our attention to the local untreated, asking how charitable care changes hospitals’ costs, revenue, and insured demand. Intuitively, this can be examined via the regression:

$$y_{ht} = \beta U_{ht} + \gamma_h + \varepsilon_{ht} \quad (5)$$

where y_{ht} is the outcome, U_{ht} is charity admissions in period t , and γ_h is a hospital fixed effect that captures the hospital's baseline. The challenge is that changes in a hospital's charitable demand likely correlate with changes in the local population, insured preferences, and potentially changes in the hospital's production technology. To address this endogeneity concern, we build an instrument based on the exposure of the local untreated to expansion spillovers, in the spirit of a Bartik instrument (Bartik, 1991; Goldsmith-Pinkham et al., 2020).

We construct the instrument by distributing a treatment-intensity measure from spillover-treated hospitals to their local untreated neighbors. We measure treatment intensity by the total charitable care spending of a hospital's owning system in expansion states in 2013. Intuitively, systems that spent more on charitable care in expansion states before the policy lost more charitable care demand and thus faced a stronger push to increase spending in non-expansion states. Panel A of Appendix Table A.2 shows this positive association between changes in charitable care and spillover treatment intensity. We distribute this intensity among the local untreated hospitals, assigning each a share inversely proportional to its distance to the spillover-treated hospital, to account for how distance affects patient substitution. Finally, the local exposure of a local untreated hospital is defined as the sum of its share of each spillover exposure.¹¹ We discuss further the instrument construction and the econometric argument for its validity in Appendix B.4.

Table 2 shows the estimates when instrumenting U_{ht} with the local exposure metric.¹² Panel A explores the effect of charitable care admissions on insured care. The estimates indicate that each charitable care admission crowds out about 8 bed-days of privately insured care and a statistically insignificant 2 bed-days of Medicare and Medicaid care. Notably, the average length of stay for charitable care admissions at local untreated hospitals is 3.77 days, which does not change with exposure to spillovers (see Appendix Table A.4). Therefore, each charitable care admission displaces more resources than it uses on average, consistent with the theory of Section 2. Unlike the theory, however, these estimates measure the effect of an additional charitable care admission rather than that of moving one potential patient from insurance to uninsurance. In addition, in reality, some hospitals offer scheduled care to those seeking charitable care, and the spillover instrument shifts both types of patients (see Appendix Table A.4). These differences rationalize why Table 2 shows a displacement of insured care from both scheduled and emergency care. Differences across insurance types reflect local insurance composition and patients' willingness to travel rather than postpone care. They are inconsistent with profit-driven selection by hospitals, as privately insured patients are more lucrative to serve than those on Medicare and Medicaid.

Panel B of Table 2 shows the estimated effect of charitable care on hospital financials. We

¹¹Formally, local exposure is defined as $LE_{ht} = \sum_{\tau \leq t} \sum_{e \in \mathcal{E}_\tau} \mathbb{1}\{5 < \text{dist}(h, h(e)) \leq 15\} \frac{1/\text{dist}(h, h(e))}{\sum_{h' \in \mathcal{H}} \mathbb{1}\{5 < \text{dist}(h', h(e)) \leq 15\} 1/\text{dist}(h', h(e))} E_{h(e)}$ where $\mathcal{E}_\tau$ is the set of expansion events occurring at period τ , $h(e)$ is the spillover-treated hospital associated with the event, $\text{dist}(h, h(e))$ is the distance between hospital h and the spillover-treated hospital, and $E_{h(e)}$ is the spillover intensity. In estimation, we also interact the instrument with indicators for whether the hospital is publicly or privately owned.

¹²See Appendix Table A.3 for the OLS estimates. Column 4 of Appendix Table A.2 shows the first-stage regression.

find that an additional charitable care admission reduces hospitals' profits by about \$29,160, with \$36,335 in lost revenue from displaced patients and an offsetting \$7,175 cost decrease. The large revenue decline is consistent with hospitals serving more charitable care patients and fewer patients overall. Hospital financial data do not clearly separate inpatient revenue by insurance source, except for Medicare, which is shown to decrease in column 3 of Panel B. Previous work, however, has established standard approaches for approximating revenue from privately insured patients (Dafny, 2009), which also shows a significant decrease in column 4.¹³ With this proxy, the estimates suggest that revenue loss is dominated by displacement of the privately insured. The estimates also indicate that labor costs at hospitals are not materially affected by charitable care admissions, consistent with hospitals reallocating resources rather than removing them.

The estimates suggest that for each additional charitable care admission, uncompensated care costs increase by \$15,793 and charitable care costs by \$13,827. Using the first number and assuming charitable care demand is driven by the uninsured, we can compute the average uncompensated cost per uninsured person as \$1,178.4 between 2011 and 2019.¹⁴ For comparison, Garthwaite et al. (2018) estimate a cost of \$830.5 per uninsured in 2011-adjusted dollars based on changes in insurance access within states and detailed financial data from 1988 to 2011. Converting to 2023 dollars yields a remarkably similar estimate: \$1,125 per uninsured.

The estimates highlight several important findings. First, consistent with Garthwaite et al. (2018), hospitals appear unable to shift the meaningful burden of charitable care to the insured. Otherwise, we would expect insurance profit to remain stable, with insured-admission revenue increasing to offset higher charitable-care costs. This contributes to an extensive literature debating the existence of such cost-shifting (or cross-subsidizing), with mixed evidence (Cutler, 1998; Frakt, 2011; White, 2013; Clemens and Gottlieb, 2017; Dranove et al., 2017). Second, we find that the opportunity cost of charitable care is sizable and likely driven by a combination of lost profit from crowded-out insured patients and from an overall decrease in admitted volume. Finally, the estimates suggest that the cost of uninsurance is high—with an estimated 29.3 million uninsured individuals in the US in 2019 (KFF, n.d.), the expected cost of their care is \$42.3 billion.¹⁵ For comparison, the realized total reported uncompensated care cost in 2019 was \$44.6 billion, inflation-adjusted to 2023 dollars (Medicaid and CHIP Payment and Access Commission, 2023). This

¹³We caution that private revenue is a proxy, discussed further in Appendix B.3. Recent work shows that these proxies have moderate cross-sectional correlations with private prices but very limited time-series correlation (Dranove et al., 2026). Medicaid revenue is reported unevenly and mixes different payment types. In our structural analysis of Florida, we meticulously reconstruct Medicaid payments using state- and year-specific payment rules.

¹⁴We compute the cost per uninsured person by taking the estimated cost of uncompensated care (including charity and bad debt) and multiplying it by the ratio of charitable care admissions to the uninsured population. Because our charitable care admissions do not cover all states, we use Florida as a representative large state and use American Community Survey data to count the uninsured population in the state.

¹⁵Using Florida's 2019 charitable admission rate, the estimated annual uncompensated care cost per uninsured person is \$1,441.5 in 2023 dollars.

suggests that our variation fairly accurately predicts uncompensated care costs.

4.5 System Targets and Spillover Mechanisms

The spillover patterns presented above raise the question of why and how systems achieve these changes. The evidence shows changes in admissions from patients residing in the same state as the affected hospitals; hence, the effects cannot be due to patient steering across state lines. Moreover, because our main evidence comes from discharge data and aggregate uninsured demand is stable at the non-expansion state level, the changes are unlikely to reflect relabeling of patients in either cost accounting or patient discharge. In addition, since the control group in our main analysis comprises other not-for-profit hospitals in non-expansion states, the effects are unlikely to be driven by local changes in hospitals' incentives to provide charitable care or by local policies unknown to us. Therefore, the main standing hypothesis is that spillover-treated hospitals simply expanded the resources available for charitable care. One plausible way hospitals can achieve this is by increasing the share of operating room and post-surgical capacity and resources reserved for emergency admissions, thereby limiting surgeons' ability to book these resources in advance for scheduled care. This mechanism requires unmet demand for charitable care at spillover-treated hospitals, allowing those hospitals to increase charitable care by increasing supply. The evidence on distance and quality discussed above supports this hypothesis.

With a theory for the mechanism behind the changes, we now ask why those occurred. As noted in Section 3, charitable care spending reported to the IRS is fairly stable over time within systems. If care expansions in spillover-treated hospitals are an attempt to retain this stability, then we should expect a few key patterns to hold. First, only systems whose spending is meaningfully affected and that can stabilize it through supply expansions should change. Intuitively, a system whose hospitals are primarily located in expansion states has a limited margin to offset a reduction in charitable care by expanding services in non-expansion states. By contrast, a system found almost entirely in a non-expansion state experiences little impact from Medicaid expansions and thus little incentive to alter its operations. Panel A of Appendix Table A.5 shows that systems in the middle of the exposure distribution expand their services the most in non-expansion states. Second, we would expect spillover systems—which can respond to Medicaid expansions by increasing charitable care in non-expansion states—to retain more of their charitable care spending than systems found only in expansion states. Panel B of Appendix Table A.5 shows that this is the case.

Although the evidence above is consistent with an attempt to maintain stable targets, it does not preclude other motives. Notably, Appendix Table A.6 shows grants and donations are positively associated with system-level charitable care spending. While these inflows are insufficient to cover charitable care spending, they may help hospitals facing financial constraints and can capture public goodwill toward the system. Thus, systems may be expanding care in non-expansion states for a variety of reasons. We note, however, that our main analysis is largely agnostic to the reason

behind these changes, as our primary identifying variation comes from the induced reallocation of charitable care demand away from the local untreated hospitals.

4.6 Discussion

Our evidence indicates that non-profit systems spanning multiple states created a channel for Medicaid expansions to spill over to non-expansion states. These spillover effects changed the behavior of the hospitals forming this link, but created demand changes that rippled across broader regions. This shift produced variation that is particularly amenable to studying uninsured care costs and regulation. Unlike variation stemming from changes in coverage—such as the direct effects of Medicaid expansions—it speaks to the costs of uninsured care in areas where this care is still prevalent. Similarly, unlike variation produced by hospital closures or broader structural changes, it is more likely to reflect the costs hospitals incur regularly. However, without further structure, our ability to leverage this variation to answer key questions about the externalities of uninsurance or the incidence of policies covering the uninsured is limited. The remainder of this article aims to close this gap by augmenting the variation with a theoretical framework that would afford us detailed quantification and counterfactual simulations, at the cost of stronger assumptions about hospital and patient behavior.

5 Model

This section develops a framework to study how charitable care burden affects resource allocation and how those changes affect the value of insuring those at the margin of charitable care. We focus on hospitals' allocation of finite resources between service lines and their management of stochastic admission requests. Service lines, such as general emergency care or scheduled oncological surgery, organize hospital services that share resources and planning. Our model is geared toward analyzing allocation and crowdouts and is parsimonious in several ways, which we discuss at the end of this section. The model is not meant to explain why spillovers occur, but rather to capture hospitals' reactions to changing demand compositions and incentives, as experienced by local untreated hospitals in Section 4. In what follows, we often refer to those seeking charitable care simply as uninsured.

We model the interaction between hospitals and patients as the subgame-perfect Nash equilibrium of a two-stage game. First, hospitals allocate resources between service lines and set rules governing patient admission. Second, patients fall ill and seek care at hospitals based on their preferences, insurance coverage, and each hospital's ability to admit them at the time of illness. These interactions occur at regular intervals (e.g., every quarter) and are independent across time and markets. Thus, we present the model below for a single market and period, omitting time t and market m subscripts. Throughout, bold lowercase letters denote vectors and calligraphic

uppercase letters denote sets, with their sizes denoted by the same non-calligraphic letter.¹⁶

5.1 Demand

Each individual i in the market belongs to a population segment $s \in \mathcal{S}$, which indexes their demographic attributes and insurance status. Members of each segment fall ill with a diagnosis $d \in \mathcal{D}$ according to a Poisson process, which is independent across segments and diagnoses.¹⁷ Conditional on falling ill, patient i seeks care among the set of hospitals available in her network $\mathcal{H}_i$. The patient's indirect utility of care at hospital $h \in \mathcal{H}_i$ is given by $v_{idh} + \epsilon_{idh}$, where v_{idh} is the systematic part of her utility and ϵ_{idh} is an idiosyncratic preference shock. In addition, the patient can seek care outside the market and obtain indirect utility of ϵ_{id0} . All idiosyncratic shocks are drawn independently from a type-1 extreme value distribution.

Independent of the patient's preferences, hospitals might reject her admission request due to capacity constraints. We denote by $\tilde{a}_{id} \in \{0, 1\}^{H_i}$ the realized acceptance vector indicating whether the patient could be admitted to each hospital at the time of her illness, and assume that the outside option always has excess capacity. Integrating over the distribution of idiosyncratic shocks and the acceptance vector, patient i is admitted to hospital $h \in \mathcal{H}_i$ conditional on illness with probability

$$\sigma_{idh} = \mathbb{E}\left[\frac{\tilde{a}_{idh}e^{v_{idh}}}{1 + \sum_{h' \in \mathcal{H}_i} \tilde{a}_{idh'}e^{v_{idh'}}}\right] = a_{idh}\mathbb{E}\left[\frac{e^{v_{idh}}}{1 + e^{v_{idh}} + \sum_{h' \in \mathcal{H}_i \setminus \{h\}} \tilde{a}_{idh'}e^{v_{idh'}}} \middle| \tilde{a}_{idh} = 1\right] \equiv a_{idh}R_{idh}, \quad (6)$$

where the first expectation is over the distribution of $\tilde{a}_{id}$, while in the second we have expanded over the two possible realizations of $\tilde{a}_{idh}$, which takes a value of 1 with probability a_{idh} . We denote by R_{idh} the probability that patient i chooses hospital h conditional on h being available, averaging over the conditional distribution of rival hospitals' availability. The choice probability of the outside option is similar, with the numerator in the expectation replaced with 1.

This structure can be microfounded in a costless search model. When the patient falls ill and draws her preferences, she and her doctor combine their knowledge of hospital states and contact everyone about whom they lack information. Assuming that the search costs are negligible relative to differences in the value of care, the patient will obtain the current state of all admitting hospitals and choose her preferred one among those available. We can thus consider the patient to require care during the period in which she falls ill, and rejections to result from hospitals offering to schedule care at an unacceptable delay. Thus, we make two key assumptions: first, that patients cannot postpone their care beyond the current period; second, that preferences are independent

¹⁶For example, x is a scalar and $\mathbf{x}$ is a vector. If X is a set, then $|X|$ is its cardinality.

¹⁷The independence assumption is ubiquitous in structural models of hospital demand, with exceptions in epidemiology and infectious disease models (Chan et al., 2016; Greenwood et al., 2019). Conditional on independence, the Poisson assumption is mild, since hospitalization is a rare event in a large population. See Appendix C.2.

of the state of the hospital, conditional on being ill in a given period.¹⁸

5.2 Supply

Each hospital h is characterized by the set of service lines it owns, $\mathcal{J}_h$, and its total resources, indexed by the total available bed-days B_h . The hospital's problem has two stages: resource allocation and admission planning. In the first stage, the hospital decides the resources to allocate among its lines to maximize its total profits:

$$\max_{b_h \geq 0} \sum_{j \in \mathcal{J}_h} \pi_{hj}(b_{hj}) \quad \text{s.t.} \quad \sum_{j \in \mathcal{J}_h} b_{hj} \leq B_h \quad (7)$$

where $\pi_{hj}(b_{hj})$ is the expected profit of line j given resources b_{hj} , with the expectation taken over its demand. Profits are subjective and may include preferences for consumer surplus.

The capacity allocation problem is well studied in hospital operations (Green, 2005; Hulshof et al., 2012). In particular, hospitals must determine how much capacity to allow surgeons to book in advance and how much to reserve for patients arriving for emergency care (Gerchak et al., 1996; Jung et al., 2019). Total planning capacity at hospitals is often rigid, as adding permanent beds requires licensing, certifications, and substantial ancillary investment in bedside resources (e.g., monitors). Moreover, in some states, hospitals must justify the public need for additional capacity and obtain regulatory approval before adding beds (National Conference of State Legislatures, 2025). Therefore, in the short run, a hospital's response to demand and profit changes is likely to begin with capacity reallocation rather than total capacity adjustment.

Given the allocated resources, the hospital must determine whom to admit. In practice, hospitals manage lines via waitlists and priority queues (Green, 2006), which pose several challenges. First, waitlists rarely appear in administrative data and are imperfectly codified and discretionary (Dixon and Siciliani, 2009; Cullis et al., 2000). Second, with waitlists, a hospital's queue affects demand at all other hospitals, creating an intractable interaction in demand across hospitals. This makes studying competition and estimation extremely challenging. Finally, even if one assumes away these issues by considering a single-firm problem or static demand, the data will largely reject any model of rigid capacity because hospitals sometimes operate above capacity (Sharma et al., 2008; Kelen et al., 2021). This is despite the rigidity of planning capacity, as hospitals can temporarily reallocate beds from one use to another, hire flexible staff, and rent beds or equipment when demand is high (Hick et al., 2004; ASPR TRACIE, 2025).¹⁹ Given these challenges, we

¹⁸The independence assumption rules out elements such as realized wait times from shaping preferences. However, expected wait times based on information available at the beginning of the period can be incorporated without loss. This contrasts with work on feedback between wait times and demand (Martin and Smith, 1999; Russo, 2024).

¹⁹Licensed bed capacity is distinct from staffed and operational bed capacity (Centers for Disease Control and Prevention, 2026). In our data, hospitals with more than 50 beds have on average 12.4% of their total licensed beds unstaffed, with a median of 5.7%. Some states permit temporary operation above licensed capacity under specified

model hospitals as choosing admission rates such that expected admitted bed-days do not exceed allocated capacity. In Appendix C.1, we show that this model is closely related to one in which the hospital manages a queue and provides wait-time guarantees to patients, treating patients first-come, first-served. Specifically, admission-rate control affords hospitals payoffs analogous to those of queue management while simplifying interactions among firms.

Formally, service line j 's admission-planning problem is to choose a rate, a_{hj} , at which to accept incoming patients. For each service line j , we define $\mathcal{K}_j = \{(s_k, d_k)\}_{k=1}^{K_j}$ as the set of population-segment and diagnosis pairs that the line serves. We denote n_{hk} as the total number of admissions from group k at hospital h .²⁰ Upon admission, each patient i of type k draws a length of stay ℓ_i^k independently from a distribution that depends on their type and has strictly positive support. We denote the total admitted length of stay as $N_{hj}^L \equiv \sum_{k \in \mathcal{K}_j} \sum_{i=1}^{n_{hk}} \ell_i^k$. The hospital line solves

$$\pi_{hj}(b_{hj}) = \max_{a \in [0,1]} \mathbb{E} \left[\sum_{k \in \mathcal{K}_j} n_{hk}(a) p_{hk} - C_{hj}(a, b_{hj}) + \tau_{hj} S_{hj}^u(a) \right] \quad \text{s.t.} \quad \underbrace{\mathbb{E}[N_{hj}^L(a)] \leq b_{hj}}_{\text{ex-ante capacity constraint}} \quad (8)$$

where p_{hk} is the revenue per admission of type k , $C_{hj}(a, b_{hj})$ is the total realized cost of the service line given its admitted demand, $S_{hj}^u(a)$ is the realized surplus of the uninsured, and $\tau_{hj} \geq 0$ is the dollar-valued weight that the hospital puts on serving the uninsured. The ex-ante capacity constraint stipulates that the line manager cannot choose a rule that would lead the service line to overflow in expectation. This allows the model to capture the binding constraints hospitals face while accounting for potential excess demand.

Hospitals with overflowing service lines might face high costs from the additional burden on staff and resources, payments to flexible staff, and rented capital. In addition, hospital efficiency may suffer when demand exceeds the hospital's physical capacity. To incorporate these costs, we model the hospital's cost function as:

$$C_{hj}(a, b) = \underbrace{\sum_{k \in \mathcal{K}_j} n_{hk}(a) c_{hk}}_{\text{surgical cost}} + \underbrace{c_{hj}^V N_{hj}^2(a)}_{\text{congestion cost}} + \underbrace{\exp(c_h^o(N_{hj}^L(a) - b))}_{\text{overflow cost}} \quad (9)$$

where $N_{hj} \equiv \sum_{k \in \mathcal{K}_j} n_{hk}$ is the total number of admissions. Above, the first term is the surgical cost paid per admission, the second captures the congestion cost, and the third the overflow cost.²¹

conditions—for example, emergency permission in California (California Department of Public Health, 2018) and an annual-average limit with reporting requirements in Maryland (Maryland Department of Health, n.d.).

²⁰We omit the line subindex since each k is associated with a unique j . A patient ill with a given condition and from a particular demographic group can be served by a single service line at each hospital. For example, a pediatric orthopedic patient cannot be cared for by an adult oncological line.

²¹The physical scale of the line is absorbed in the cost parameter c_{hj}^V . The overflow cost also introduces a slight

We assume that the firm's cost parameters (c_{hk}, c_{hj}^V, c_h^o) are positive and note that the overflow cost parameter is common across lines for the hospital, consistent with it emerging from a common cost of rental and staffing that grows increasingly expensive as a line overflows.

Given the non-discriminatory nature of admission rates and the simultaneity of hospital choices, we can simplify the firm's problem. In particular, since each individual-condition pair (i, d) is associated with a single consumer type k , we can write the conditional choice probability in Equation (6) as $\sigma_{kh} = a_{hj(k)} R_{hk}$, where $j(k)$ is the service line associated with k . In addition, each admission rate is independent of the others, conditional on the information available at the beginning of the period. Hence, the request probability, R_{hk} , is not a function of hospital h 's admission rate. Letting λ_k denote the rate at which type k patients fall ill, properties of the underlying Poisson process imply that Equation (8) simplifies to (proof in Appendix C.3):

$$\begin{aligned} \pi_{hj}(b_{hj}) = \max_{a \in [0,1]} & a \left(\sum_{k \in \mathcal{K}_j} \lambda_k R_{hk} (p_{hk} - c_{hk}) - c_{hj}^V \Lambda_{hj} + \tau_{hj} \Delta S_{hj}^u \right) - a^2 c_{hj}^V \Lambda_{hj}^2 - \exp(a \chi_{hj} - c_h^o b_{hj}) \quad (10) \\ \text{s.t.} \quad & a \Lambda_{hj} \mu_{hj}^L \leq b_{hj} \end{aligned}$$

where $\Lambda_{hj} \equiv \sum_{k \in \mathcal{K}_j} R_{hk} \lambda_k$ is the maximum expected admissions, ΔS_{hj}^u is the line's expected contribution to uninsured consumer surplus, μ_{hj}^L is the mean length of stay among patients admitted to line j , and $\chi_{hj} = \Lambda_{hj} (M_{hj}^L(c_h^o) - 1)$ depends on the overflow cost parameter, the maximum expected admissions, and the line's length-of-stay distribution.²² Here $M_{hj}^L(t)$ is the moment-generating function of the length of stay of a patient admitted to line j .

The following proposition, proven in Appendix C.4, characterizes the solution to the hospital's resource allocation and acceptance rate problem. In what follows, we denote by $a_{hj}^o(b_{hj})$ and $a_{hj}^*(b_{hj})$ the unconstrained and constrained solutions to Equation (10), respectively. We denote (a_{hj}^*, b_{hj}^*) as the solutions to the hospital's overarching problem.

Proposition 1. *Suppose that for every hospital h and line $j \in \mathcal{J}_h$, $\Lambda_{hj} > 0$ at every profile of rivals' admission rates, $\mathbb{E}[e^{c_h^o i}] < \infty$ for every $k \in \mathcal{K}_j$, and $\mathbb{E}_i[\log(1 + e^{v_{ih}})] < \infty$ for every $k \in \mathcal{K}_j^u$. Then $a_{hj}^o(b_{hj})$ and $a_{hj}^*(b_{hj})$ have closed-form expressions and satisfy the following properties: (i) Conditional on rivals' admission rates, the hospital's problem is jointly strictly concave in $(\mathbf{a}_h, \mathbf{b}_h)$; (ii) $a_{hj}^*(b_{hj})$ is strictly increasing in b_{hj} whenever $a_{hj}^*(b_{hj}) \in (0, 1)$; (iii) For any $j, j' \in \mathcal{J}_h$ with $b_{hj}^*, b_{hj'}^* > 0$, $\pi'_{hj}(b_{hj}^*) = \pi'_{hj'}(b_{hj'}^*)$ whenever both derivatives exist; (iv) The game admits an equilibrium in pure strategies $(\mathbf{a}^*, \mathbf{b}^*)$.*

We rely on this proposition extensively in our analysis. The assumptions establish basic fea-

under-utilization cost. The true, intractable, overflow cost of a hospital can be modeled as $\exp(c_h^o \max\{N_{hj}^L(a) - b, 0\}) - 1$. The constant subtraction does not alter the firm's planning and is therefore ignored in the analysis. The absence of a kink at zero is an important simplification but is of minimal numerical relevance given the exponential decay.

²²Formally, $\Delta S_{hj}^u = \sum_{k \in \mathcal{K}_j^u} \lambda_k \mathbb{E}_{i, \mathbf{a}_{-h}} \left[\ln \left(\frac{1 + e^{v_{ih}} + \sum_{h' \in \mathcal{H}_i \setminus \{h\}} \bar{a}_{h'i} e^{v_{ih'}}}{1 + \sum_{h' \in \mathcal{H}_i \setminus \{h\}} \bar{a}_{h'i} e^{v_{ih'}}} \right) \right]$ where $\mathcal{K}_j^u$ is the subset of patient types that are uninsured.

sibility, guaranteeing the hospital line a positive and finite demand and a manageable overflow risk.²³ The closed-form expression for $a_{hj}^*(b_{hj})$ improves the tractability of the model's analysis and simulation. The first part of the proposition is fundamental, as it implies that each hospital's problem has a unique solution. The second establishes that admission rates can be inverted to recover resource allocations, and the third states that, because resources are transferable across service lines, the manager must be indifferent in allocating the marginal unit at the optimum. Together, these will prove crucial for identification and estimation, allowing us to identify differences in marginal costs across lines through differences in their marginal revenues. Finally, the fourth condition ensures that the game admits an equilibrium, which might not occur if the constraint on hospital actions generated discontinuities in the best-response mapping across firms. In the proof appendix, we also discuss conditions for equilibrium uniqueness, which require assuming upper bounds on admission rates and strategic interactions. Intuitively, this would happen if hospitals are sufficiently differentiated. However, establishing economically meaningful global conditions for uniqueness in oligopolistic games is a common challenge shared by our framework.

5.3 Discussion

The model above presents a tractable framework for studying resource allocation and crowdout in hospital care. Service lines form the basis for product segmentation and production in the model, with demand and resources assigned to one line being isolated from those of others. While service lines are organizational structures used by hospitals in practice, our model assumes a few key simplifications. First, admission rates are non-discriminatory within each service line; therefore, any patient selection occurs only across lines. Thus, patients are selected by allocating fewer resources to less profitable lines, but patients sharing a service line must be treated equally. Second, the binding resource is bed-days, which implicitly assumes that labor is flexible and that hospitals have sufficient fixed capital to treat patients across lines, or can rent it when needed. Third, because resources are line-specific, overflows resolved ex post by reallocating idle resources within the hospital carry the same opportunity cost as overflows resolved with external resources.²⁴

Our model fits within the workhorse structural insurer-hospital framework (Town and Vistnes, 2001; Gowrisankaran et al., 2015; Ho and Lee, 2017). As long as insurer-hospital contracts do not condition on resource allocations, the framework can easily accommodate a pricing stage. We do not model pricing here because our data are not amenable to studying hospital-insurer negotiations, and we expect changes in uninsured care to have minimal impact on these prices.

²³Without the first assumption, the line might lack demand; without the second, overflow risk would force it to shut down; and without the third, patient expected surplus is unbounded. A sufficient condition for positive demand is that at least one consumer type k has finite preferences and positive illness probabilities. The second requires that length of stay decay fast enough relative to overflow. For the third, it suffices that the indirect utility distribution be finite over finite utilities or sufficiently thin-tailed.

²⁴A sufficient condition is that hospitals be able to lease idle capacity at the same market rate at which they rent it.

Our demand model generalizes the standard logit demand model used in the literature but is more rigid than a general consideration-set model (Abaluck and Adams-Prassl, 2021; Barseghyan et al., 2021; Agarwal and Somaini, 2026). In particular, we assume patients know all relevant hospitals, so admission events arise exclusively from preferences and availability. This entails a trade-off between generality and tractability: we show below that our demand model yields a simple choice likelihood, whereas general consideration-set models are often hard to estimate and simulate. In addition, we assume that a patient’s length of stay is beyond the hospital’s control despite evidence that hospitals have a say over discharge timing (Doyle, 2005). If changes in the uninsured burden and the compensating policies we study do not meaningfully alter the speed at which hospitals discharge each consumer type, this assumption is likely innocuous. In exchange, our model provides a tractable framework for competition with binding capacity constraints that can rationalize crowdouts, resource reallocation, and the patterns of Section 4.

6 Identification and Estimation

We estimate the model using individual-level data from Florida for 2010 to 2019, with quarters as the period length. We adopt a parsimonious classification of service lines, splitting them into emergency care, scheduled insured care, and scheduled uninsured care. This classification allows hospitals to satisfy emergency care regulations and divide scheduled care along the insurance lines we are interested in. Scheduled insured surgery does not discriminate in admissions across insurance types, but we allow hospitals to reject Medicaid or Medicare patients by excluding those hospitals from patients’ choice sets.²⁵ Proofs and technical details are relegated to Appendix D.

6.1 Demand

Our demand model and identifying variation are a special case of the setting considered in Agarwal and Somaini (2026), and are almost identical in structure to the empirical application in that article. As established there, separately identifying patient preferences from acceptance probabilities requires a shifter of preferences excluded from hospitals’ acceptance decisions, and a shifter of acceptance decisions excluded from patients’ preferences. The first shifter is naturally the distance from a patient’s home to the hospital, which affects patients’ preferences for hospitals but—all else equal—is irrelevant for hospitals’ admission-rate decisions. The second shifter is expected congestion at the hospital, which shifts hospitals’ chosen admission rates, but is assumed to be excluded from patients’ preferences. In particular, in our application, hospital congestion changes due to Medicaid spillover effects and disease seasonality.

We estimate the model via a nested fixed-point maximum-likelihood procedure. In Ap-

²⁵In particular, we assume that hospitals without an emergency room are excluded from emergency patients’ choice sets. For scheduled care patients, we assume Medicaid patients can access only hospitals that have historically accepted Medicaid scheduled surgery, and similarly for Medicare patients.

pendix D.1 we show that patient i 's conditional choice probabilities can be written as

$$\sigma_{idht} = a_{hj(i,d)t} \bar{\sigma}_{idht} \int_0^\infty e^{-u} \prod_{h' \in \mathcal{H}_i \setminus \{h\}} \left(a_{h'j(i,d)t} + (1 - a_{h'j(i,d)t}) e^{u \bar{\sigma}_{idh't}} \right) du \quad (11)$$

where $\bar{\sigma}_{idht} = e^{v_{idht}} / (1 + \sum_{h' \in \mathcal{H}_i} e^{v_{idh't}})$ is the standard logit choice probability without admission constraints. If all hospitals accept patient i with probability one, then Equation (11) reduces to the standard logit form. In contrast, admission probabilities increase with rivals' rejection rates, as governed by the last term in the integral. This expression allows us to evaluate the data log-likelihood given a parametrization of patients' preferences, v_{idh} , and a vector of acceptance probabilities, $\mathbf{a}$. In estimation, we recover acceptance probabilities from the fixed-point contraction map, $\ln(a_{hjt}) = \ln(N_{hjt}^{L*}) - \ln(\sum_{i \in \mathcal{I}_{jt}^*} R_{iht}(\mathbf{a}, \mathbf{v}) \ell_i^*)$, where N_{hjt}^{L*} is the observed total admitted length of stay at the hospital, $\mathcal{I}_{jt}^*$ is the observed set of individuals seeking admission for line j in period t , $R_{iht}(\mathbf{a}, \mathbf{v})$ is the model-implied request likelihood, and ℓ_i^* is the patient's observed length of stay.

We model patients' preferences as a linear function of distance and systematic preferences for hospital-line-diagnosis-group combinations. In particular, $v_{idh} = \alpha_{i,j(i,d)} \text{dist}_{ih} + \gamma_{j(i,d)dh}$, where dist_{ih} is the distance from a patient's home to the hospital, and $\gamma_{j(i,d)dh}$ is a hospital line-by-diagnosis-group fixed effect. We allow patients' disutility of travel to vary by service line, market, gender, age group, and whether they live in a ZIP code with a mean income below the state median.²⁶ We define markets separately for each service line by grouping contiguous hospital referral regions (HRRs) so that at least 90% of demand originating in each market is served within that market.²⁷ Appendix Figure A.2 shows the resulting market classifications, highlighting how emergency markets are narrower given that patients usually seek care closer to home, while uninsured scheduled care markets are more extensive. In estimation, we normalize patients' distance to their in-network hospitals by their distance to their nearest out-of-market provider to account for differences in the cost of substituting local for out-of-market care.

6.2 Supply

Identifying hospitals' costs hinges on the optimality of their admission and resource allocation decisions. From the econometrician's perspective, we can consider the hospital's revenue, admissions, length of stay, and total available bed-days as data. Given the identified demand, we also treat admission rates as known. However, resource allocation across lines is unobserved.

²⁶Age groups follow HCUP definitions: under 6, 6 to 17, 18 to 24, and then in ten-year increments until age 74. Ages 75 and over are a single category. We define diagnostic groups by selecting the 30 most common DRG codes for each age group and gender and clustering the remaining codes into 10 bins based on their DRG weights.

²⁷We adopt this broad market classification for two reasons. First, uninsured scheduled care is often delivered far from home, likely due to the scarcity of providers willing to offer those services. Hence, standard market classifications used in the literature are unlikely to be appropriate for this population. Second, we need to minimize the share of patients allocated to the abstract out-of-market provider to maintain accurate capacity utilization accounting in counterfactuals.

Proposition 1(ii) provides the remedy: resource allocations may be inverted from admission rates.

Before stating the main identification result, we add two assumptions on the hospital's costs and present the key identifying conditions. First, we assume surgical costs are an affine function of the observed resource intensity measure (i.e., DRG weights), ω_k , such that $c_{hkt} = \omega_k c_{hjt}^S + c_{hjt}^F$.²⁸ Second, we assume that a hospital's cost parameters do not vary within a year, implying that we observe the hospital making four choices—one per quarter—under the same cost configuration, but potentially different demand. Let $\theta_{hjt} = (c_{hjt}^F, c_{hjt}^S, c_{hjt}^V, \tau_{hjt})$ collect the linear cost and weight terms. The first-order conditions for the optimal admission rate of Equation (10) and the optimal resource allocation of Equation (7) state that for any line with $a_{hjt} \in (0, 1)$ it holds that:

$$p_{hjt} = \underbrace{x'_{hjt} \theta_{hjt}}_{\text{marginal service cost}} + \underbrace{\frac{\chi_{hjt}(c_{ht}^o)}{\Lambda_{hjt}} \exp(\chi_{hjt}(c_{ht}^o) a_{hjt} - c_{ht}^o b_{hjt}) + v_{hjt} \mu_{hjt}^L}_{\text{marginal capacity cost of admission}} \quad (12)$$

$$\eta_{ht} = \underbrace{c_{ht}^o \exp(\chi_{hjt}(c_{ht}^o) a_{hjt} - c_{ht}^o b_{hjt}) + v_{hjt}}_{\text{marginal value of allocated capacity}} \quad (13)$$

where $p_{hjt} := \sum_{k \in \mathcal{K}_j} \frac{\lambda_{kt} R_{hkt}}{\Lambda_{hjt}} p_{hkt}$ is the expected revenue of the marginal admission to the line, and the term $x'_{hjt} \theta_{hjt}$ collects surgical and congestion costs, where $x'_{hjt} := (1, \sum_{k \in \mathcal{K}_j} \frac{\lambda_{kt} R_{hkt}}{\Lambda_{hjt}} \omega_k, 1 + 2\Lambda_{hjt} a_{hjt}, -\frac{\Delta S_{hjt}^u}{\Lambda_{hjt}})$ is a vector of simple aggregate elements: a constant, the expected resource intensity of an admission to the line, one plus twice the expected admissions, and the average effect of admission on uninsured surplus. The term labeled *marginal capacity cost of admission* is the sum of the marginal admission's effect on overflow risk and the opportunity cost of its expected bed-day use. In particular, v_{hjt} is the Lagrange multiplier associated with the admission-rate constraint and measures the within-line opportunity cost of an admission: it is zero if, in expectation, the line has idle capacity, and positive if the marginal patient crowds out a profitable one. Thus, Equation (12) simply states that the revenue of the marginal admission must equal its cost.

The optimality of resource allocation in Equation (13) states that the opportunity cost of an allocated bed-day, η_{ht} , must equal its marginal value at its allocated line. This relates to Proposition 1(iii), which ties the marginal profitability of resources across lines. This implies a unique internal price for a bed-day, recoverable from the optimality of the resource allocation. This optimality pins down the marginal capacity cost of an admission, leaving Equation (12) with its core linear structure. Thus, one might expect the main additional assumption needed for identification to be sufficient local variation in the components driving x_{hj} , as in a linear model.

In Appendix D.4, we show this is the case and formally prove the identification of (θ, c_h^o) ,

²⁸This structural assumption is commonplace in the literature on hospital competition (Gowrisankaran et al., 2015; Ho and Lee, 2017; Liu, 2022; Cuesta et al., 2024; Arnold et al., 2026).

hospital by hospital. The proof is extensive because the marginal capacity cost is unobserved and Equation (12) has a linear-exponential structure. It shows that Equation (13) determines the internal cost of capacity up to the overflow cost parameter c_{ht}^o . This removes the unobserved allocation from the identifying equation and replaces it with a single unobserved price. This yields a linear equation structure, with an added exponential hospital-specific cost. A technical lemma then proves that sufficient local variation in expected marginal revenue and the components driving x_{hjt} is sufficient to identify the model. Crucially, this includes variation in demand among those seeking charitable care, as provided by the Medicaid spillovers we leverage. Without it, we cannot identify either the cost of charitable care or the hospital's preference for uninsured surplus.²⁹

We estimate the model using a maximum-likelihood approach that draws on key insights from our identification argument.³⁰ It addresses the unobservability of b_{hjt} by using Equation (13) and the complementarity conditions imposed by the admission rate problem. This, however, introduces a challenge, as complementarity problems are non-convex. In addition, to leverage more information from our finite data, we cannot rely only on cases where $a_{hjt} \in (0, 1)$, as in the identification argument. We address these issues by estimating the shadow price of capacity, using a vanishing relaxation of the complementarity condition, and introducing a structural cost error with a distributional assumption to leverage the corner cases.

In particular, we introduce a parameter $\rho > 0$, and denote the slack of a service line as $s_{hjt} := b_{hjt} - \Lambda_{hjt} \mu_{hjt}^L a_{hjt}$. With this, we write the relaxed complementarity condition of line (h, j) in period t as $s_{hjt} v_{hjt} = \rho$. Clearly, as $\rho \rightarrow 0$, the condition converges to the Kuhn-Tucker complementarity condition, which holds at the optimum. Using these elements, we can write Equation (13) as

$$\eta_{ht} = c_{ht}^o \exp \left((\chi_{hjt}(c_{ht}^o) - c_{ht}^o \Lambda_{hjt} \mu_{hjt}^L) a_{hjt} - c_{ht}^o s_{hjt} \right) + \frac{\rho}{s_{hjt}} \quad (14)$$

which has a unique solution for $s_{hjt}(\rho, \eta_{ht}, c_{ht}^o)$. From this and the total capacity constraint of the hospital, we have that η_{ht} is the unique value that satisfies $\sum_{j \in \mathcal{J}_h} \left(\Lambda_{hjt} \mu_{hjt}^L a_{hjt} + s_{hjt}(\rho, \eta_{ht}, c_{ht}^o) \right) = B_{ht}$. This implies that, for a given (ρ, c_h^o) , resource slackness and the shadow cost of capacity are uniquely determined, as in the identification argument. Finally, we introduce a structural error by rewriting $c_{hjt}^F = \bar{c}_{hj}^F + \varepsilon_{hjt}$. Given that c_{hjt}^F enters Equation (12) as an additive term, the structural error enters additively as well, yielding a non-linear regression structure:

$$p_{hjt} = \mathbf{x}_{hjt}' \tilde{\boldsymbol{\theta}}_{hjt} + \psi_{hjt}(c_{ht}^o; \rho) + \varepsilon_{hjt} \quad (15)$$

where $\tilde{\boldsymbol{\theta}}_{hjt}$ is the set of cost parameters $\boldsymbol{\theta}$ with the decomposition of c_{hjt}^F applied, and $\psi_{hjt}(c_{ht}^o; \rho)$

²⁹The argument shows that the econometrician need not know the hospital's payment per insurance source. This offers a tool for research relying on one of the many data sources that do not include private insurance identifiers.

³⁰We estimate the supply model using Florida data starting in 2013, as it requires private insurance prices which are only available to us starting this year.

collects overflow and opportunity costs. We assume that structural cost shocks are normally distributed with a common variance and are drawn independently across periods and lines, slightly relaxing the assumption of yearly-homogeneous cost structures. This distributional assumption allows us to leverage the inequalities imposed by the optimality of corner solutions $a_{hjt} \in \{0, 1\}$. With this, we can write the log-likelihood of each hospital line in the data and solve³¹

$$\lim_{\rho \rightarrow 0} \max_{\theta, c^o} \sum_{h,j,t} \mathcal{L}_{hjt}(\theta, c_{ht}^o; \rho). \quad (16)$$

We solve this problem over a decaying sequence of relaxation parameters. In estimation, we allow the surgical cost slope (c_{hjt}^S) to vary by hospital-year, the fixed surgical cost ($\bar{c}_{hj}^F$) by service line type, and the congestion cost (c_{hjt}^V) by hospital-line. We allow the overflow cost parameter (c_{ht}^o) to vary by year and HRR to capture local differences in rental and labor costs. Finally, we allow hospitals' preference for uninsured surplus (τ_{hjt}) to vary by hospital system and to respond to Medicaid spillover effects. Appendix D.5 presents further details on the estimator.

7 Results

7.1 Demand

The demand estimates reveal substantial heterogeneity in acceptance rates, travel disutility, and hospital preferences.³² Figure 5a shows the distribution of hospital acceptance rates by service line. Conditional on having the appropriate service line, the median hospital accepts over 85.9% of emergency care admission requests, compared with 70.1% for scheduled insured care and 52.6% for charitable scheduled care. To relate these acceptance rates to preferences, the figure also shows how often patients are admitted to their (stochastically) most preferred hospital. The gap between the distributions indicates that although acceptance rates for uninsured scheduled patients are low, these patients favor larger public hospitals with higher acceptance rates, resulting in lower realized rejection rates. Appendix Figure A.4 presents the geographic distribution of acceptance probabilities and their connection to local travel elasticities.

The estimates reveal admission patterns consistent with our supply theory. In Appendix Figure A.5a, we show that patients in areas that experienced relative increases in uninsurance rates faced a decrease in their probability of admission to their preferred provider for emergency care and an increase in their probability of admission for scheduled care. The estimates show that crowdouts are material, but do not reveal whether they stem from capacity constraints or from misallocation due to a lack of market coordination mechanisms. In particular, conditional on insurance status and medical need, whether one patient is admitted while the other is referred to

³¹Cuesta et al. (2024) use a similar approach to address the non-convexity of a combinatorial plan-design problem.

³²Appendix Table A.7 displays distance disutility estimates. Appendix Figure A.3 displays in- and out-of-sample fit.

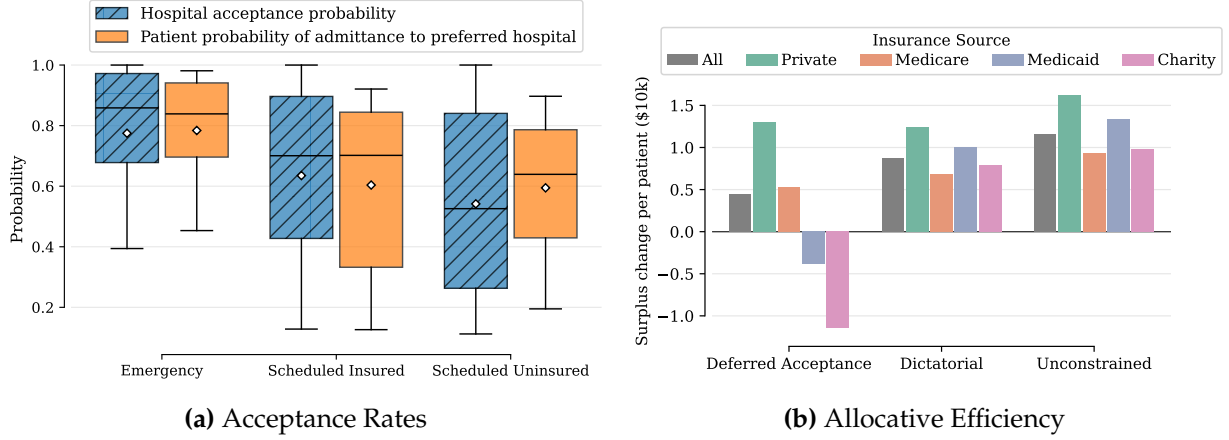


Figure 5: Acceptance Rate and Allocative Efficiency Holding Supply Fixed

Notes: Figure 5a shows the estimated distribution of acceptance rates (a_{hjt}) from the demand model. The whiskers on the box plot cover the 10th to 90th percentiles, while the box covers the interquartile range. A horizontal line represents the median, and a white diamond represents the mean. Hatched blue boxes show the distribution across hospitals. Solid orange boxes show admission probability to patients' most preferred hospital ($\sum_{h \in \mathcal{H}_i} a_{hj(i,d)t} \bar{o}_{idht} + (1 - \sum_{h \in \mathcal{H}_i} \bar{o}_{idht})$), including the outside option. Figure 5b displays the estimated allocative efficiency of the 2019 admissions relative to patient-proposed deferred acceptance matching, dictatorial surplus-maximizing assignment, and unconstrained surplus-maximizing assignment. The first two assignments are constrained to use no more of each hospital line's resources than in the status quo. All changes are relative to the status quo and allocate patients and beds within each quarter. Surplus transformed at \$1,033 dollars per mile, following Prager (2020) estimates.

alternatives often depends more on the randomness of illness onset than on preferences. Patients cannot trade their allocations, and rigid hospital prices combined with insurance coverage make prices a negligible sorting force. Therefore, efficient unrealized trade might exist between patients. To measure this, we conduct a three-step counterfactual assignment exercise: first, we introduce a market allocation mechanism; second, we solve for the patient-optimal dictatorial assignment; and finally, we measure the surplus that would result if each patient obtained access to their most preferred hospital. The first step reveals the loss from the lack of a coordination mechanism. The difference between this and the dictatorial assignment reveals the misallocation that a simple market mechanism cannot remedy. The final step reveals how capacity limits surplus. Because our model yields patient surplus as compensating variation in miles traveled for care, we dollarize surplus using estimates from Prager (2020).³³

To approximate a market-based allocation mechanism, we use a patient-proposed Deferred Acceptance (DA) matching mechanism.³⁴ Similar mechanisms often allocate scarce resources among

³³Prager (2020) estimates patients' dollar-valued disutility from travel in a setting with simple, deterministic, and known copays for hospital care. This observability and simplicity make these estimates particularly credible. To be conservative, we use the lower estimate from Prager's preferred specification, \$1,033 per patient-mile (adjusted to 2023 dollars using the CPI). We discuss alternative measures in Appendix E.1. The rankings of allocations, the signs of their surplus effects, and comparisons expressed as shares are independent of the conversion rate.

³⁴Ideally, we would simulate market-based allocations in a counterfactual with direct pricing to consumers. However, patient price elasticity is known to be modest because of factors that might not persist if patients had to pay for care directly. Nevertheless, an extensive theoretical literature draws tight connections between stable matching mechanisms and price equilibrium (Hatfield and Milgrom, 2005; Bulow and Levin, 2006; Hatfield et al., 2013; Azevedo and Leshno, 2016), justifying our use of DA as an approximation.

heterogeneous agents when prices are unavailable or undesirable, such as in labor markets (Roth and Peranson, 1999), education (Arteaga et al., 2022), the military (Greenberg et al., 2024; Davis et al., 2026), and refugee settlement (Delacrétaz et al., 2023). Intuitively, a similar system could be implemented in hospital care through a shared state-scheduling platform. These mechanisms consider patient and hospital preferences, are pairwise stable, yield the unique patient-optimal stable match, and make truth-telling optimal for patients. Figure 5b displays the 2019 estimates, revealing clear winners and losers from the absence of strong coordination in current markets. Relative to market-based coordination outcomes, privately insured patients lose \$12,914.5 per patient per year, Medicare patients lose \$5,256.9, while Medicaid and charitable care patients gain \$3,757.6 and \$11,332. This quantifies the protection the existing market arrangements afford the most vulnerable segment of the population, with Appendix Figure A.5b showing that it translates to more travel for the insured and significantly less so for the uninsured.³⁵

Dictatorial assignment shows that a heavy-handed assignment policy could increase all patients' surplus by an average of \$8,721.6 per patient per year. Some of these allocations, however, are pairwise unstable, with some patients and hospitals preferring to deviate and match outside the system if allowed. Finally, the unconstrained solution shows that average patient surplus would increase by \$11,546.2 per year if we removed capacity constraints. This shows that even under optimal allocation, 24.5% of the surplus gains would remain unattainable. Thus, adjusting for their relative probability of illness, we estimate that the average privately insured Floridian loses \$790 per year from lack of market coordination—1.09 times the average monthly marketplace premium in Florida in 2019—and up to \$991 from limited capacity. In contrast, the average uninsured Floridian gains \$1,081 from the lack of market coordination, and would benefit by \$932 if they could be admitted to their most preferred hospital without capacity or insurance constraints.

7.2 Supply

The supply estimates reveal stark differences in markups across payors.³⁶ Because our model's structure and identification argument differ from those in the literature we build on, we begin by benchmarking our estimates against prior work. Figure 6a shows how the distribution of cost-to-charge ratios—estimated annual hospital costs divided by reported total charges—compares with the estimates of Bai and Anderson (2016). While the reference computes these ratios directly from 2013 financial data, our estimator infers them from optimal resource allocation using no supplemental cost data. Yet, both yield similar estimates, with costs at 16.7% of charges for for-profit hospitals (compared to 15.8%) and 28.7% for non-profit hospitals (compared to 26.4%). In Appendix Figure A.7a, we compare our estimated markups from private insurers to a collection of

³⁵Appendix Figure A.6 also shows that moving toward DA would push a considerable share of emergency charitable care patients to out-of-market providers.

³⁶See Appendix Figure A.8 for the estimated hospital cost components. See Appendix Figure A.9 for fit.

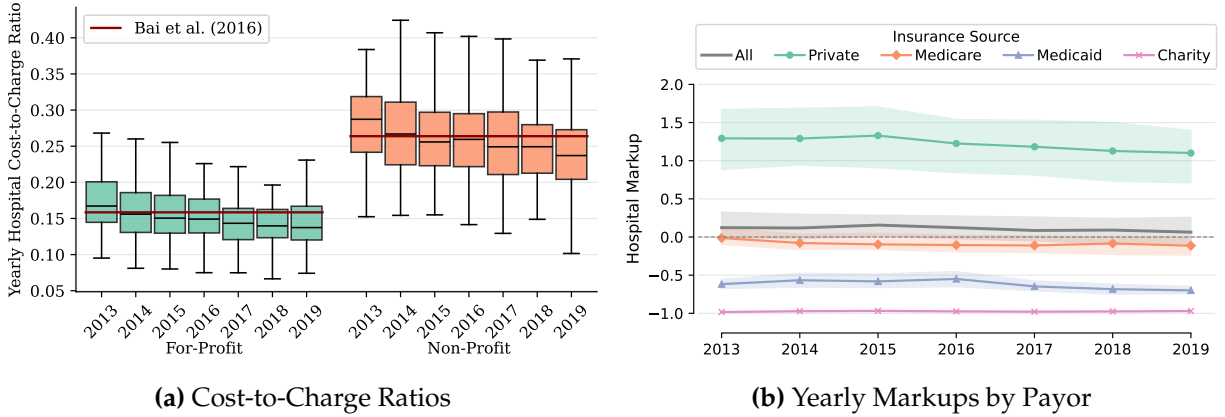


Figure 6: Estimated Markups

Notes: These figures display markup statistics implied by the structural cost estimates. Figure 6a shows hospital cost-to-charge ratio distributions, where costs are estimated, and charges are as reported by hospitals. Figure 6b shows hospitals' estimated markups over cost, measured as profit divided by total cost. Lines are admissions-weighted medians and shaded areas represent interquartile ranges across hospitals for each payor and year. See Appendix Figure A.7 for additional evidence on markups.

reference papers. Our estimates lie within the range of markups implied by the structural models of Gowrisankaran et al. (2015), Ho and Lee (2017), and Setzler (2025).

Figure 6b shows how hospital markups differ across payment sources and vary over time. We find that Medicaid pays hospitals far below their costs, while Medicare pays very close to hospital costs.³⁷ Median hospital markups over cost for Medicare patients have declined, from -1.3% in 2013 to -11.4% by 2019. Similarly, the Medicare Payment Advisory Commission's (MedPAC) analysis of hospitals' financial data found that short-term acute-care hospitals experienced declining profit margins on Medicare services (including, but not limited to, inpatient hospitalizations), reaching -8.7% in 2019 (Medicare Payment Advisory Commission, 2015, 2021). In contrast, we estimate that private insurers pay hospitals a median markup over cost of 129.3% in 2013 and 110.1% in 2019, consistent with prior studies (Lopez et al., 2020). Hospitals' overall markup is limited and declining, with the median markup at 12.3% in 2013 and falling to 6.3% by 2019. For comparison, MedPAC estimated a 7.6% all-payer margin for inpatient hospitals for 2019. The evidence on these trends in overall hospital markups contributes to the debate on the evolution of markups in the US economy (Basu, 2019; Berry et al., 2019; De Loecker et al., 2020; Grieco et al., 2024; Shapiro and Yurukoglu, 2026), as hospital care constitutes over five percent of gross domestic product (Centers for Medicare & Medicaid Services, 2026b). Particularly relevant is the concurrent work by De Loecker and Fleitas (2026), which estimates hospital markups using a production function approach; Appendix Figure A.7b shows that our all-payer margins are broadly consistent.

The estimates are consistent with the evidence from Medicaid spillovers. Our estimated cost per uninsured admission occurring between 2014 and 2019 is $\$12,406.8$, and $\$12,901.4$ when restricted

³⁷Hospital revenue includes all payments received from Medicare and Medicaid for serving low-income or uninsured patients, plus an approximate recovery rate from charitable care patients, which we obtain from hospitals' tax returns.

to hospitals experiencing spillovers.³⁸ Our estimates reveal that the median hospital values its impact on uninsured surplus at about 46% of its average revenue from insured care (see Appendix Figure A.10). Non-profit hospitals place a weight on uninsured surplus that is, on average, 27.5% higher than the weight placed by for-profit hospitals, while public hospitals place a weight that is 39.6% higher. We also find that Medicaid spillovers increased this weight by 8.9% among spillover-treated hospitals, rationalizing affected hospital behavior.

The estimates allow us to recover hospitals' implied resource allocation across lines, rationalizing the demand estimates. We find that the median hospital allocates 62% of its resources to emergency care, while uninsured scheduled care receives only 11% (see Appendix Figure A.11). Weighted by acceptance rate, only 3% of capacity is available for uninsured scheduled care, with public hospitals supplying the largest share of charitable care capacity overall. Only 12% of hospitals have at least one line at capacity in expectation. However, fewer than 5% of hospitals have all lines at capacity, suggesting that most limit acceptance rates to control costs. Therefore, changing hospital compensation for charitable care may affect aggregate capacity supply.

8 Counterfactual Policy and Externalities

Using the model, we explore the impact of alternative supply-side policies. First, we study those that compensate hospitals directly for each charitable admission, in the spirit of existing public insurance programs. Because we focus only on the payment aspect of these policies and ignore risk protection and network changes, we refer to these as pseudo-insurance policies. Second, we explore policies that strengthen non-profit hospitals' obligation to provide charity care by establishing an optimal minimum target and a market for uninsured-care credit trade.

This policy analysis has three goals. First, the pseudo-insurance policies reveal how payment differences across patients generate resource externalities, connecting directly to the economics of Section 2. Second, the credit-trading policies reveal where charitable care resources are in deficit and where tax-exempt benefits are in excess. They provide a simple metric to quantify the heterogeneity underlying the tax-to-benefit dispersion shown in Section 3.1. Finally, both policy exercises let us benchmark the externalities of uninsurance against plausible alternatives, such as restructuring DSH payment schemes, expanding Medicaid coverage, or establishing minimum requirements for hospitals, all of which have been implemented in one form or another.

The following analyses cover the last year and quarter in our sample (2019, Q4). Each policy exercise involves solving counterfactual equilibria, which entails finding a new set of acceptance rates, resource allocations, and patient request probabilities that satisfy each agent's best-response

³⁸Our structural estimates are also consistent with the estimates of Section 4 regarding the limits of cost-shifting. Appendix Figure A.7c shows the slope from regressing hospital margins on their charitable care demand share, conditional on hospital fixed effects. For every percentage-point increase in a hospital's charitable care share, its profit margin declines by 0.74 percentage points, suggesting that cost-shifting, if present, is very limited.

criterion. Throughout, we assume public hospitals are unresponsive to these policy changes and exclude them, meaning their resource allocations and acceptance rates remain at the status quo. This assumption captures public hospital behavior as governed by public authorities, which may respond more slowly, pursue different objectives, or exclude these hospitals from the binding requirements of these policies for political reasons. We present robustness to this assumption in Appendix E.4, showing that relaxing public-hospital participation yields nearly identical key findings to those reported below. Appendix E.5 provides additional implementation details.

8.1 Pseudo-Insurance Programs

Pseudo-insurance programs compensate hospitals directly for charitable care, without forming a formal insurance system for the uninsured or incorporating them into existing structures. By adjusting the payment, the regulator can influence hospitals' acceptance rate and resource allocation decisions, and thus alter all patients' access to care. We consider the case of a regulator that chooses a payment z^u per *complexity* unit (DRG weight) of admitted uninsured demand served, and seeks to maximize a weighted welfare criterion:

$$W(z^u; \gamma^u, \gamma^\pi) = \underbrace{\gamma^u S_t(a^*(z^u))}_{\text{patient surplus}} + \underbrace{\gamma^\pi \sum_{h,j} \pi_{hjt}(a^*(z^u), b^*(z^u))}_{\text{hospital profits}} - \underbrace{\sum_{h,j} \sum_{k \in \mathcal{K}_j^u} \lambda_k R_{hkt} a_{hjt}^*(z^u) \omega_k z^u}_{\text{fiscal cost}}. \quad (17)$$

In this objective, patient surplus $S_t(\cdot)$ is measured in miles, dollarized, and weighted relative to fiscal cost by the parameter γ^u . Hospital profits are weighted relative to fiscal costs by the parameter γ^π . The final term is the policy's expected fiscal cost. By paying per unit of complexity, the policy aligns with Medicare payment structures and links payments to hospitals' surgical costs.

We begin by exploring the policy's impact and optimality when surplus is dollarized and weighted at \$1,033 per equivalent mile of travel, as in Section 7.1. Figure 7a shows how consumer surplus and hospital profits net of fiscal costs change as policy payments increase. Net profits are negative and decline as government transfers to hospitals cancel out, leaving only the incremental cost of service. In contrast, as the policy becomes more generous, consumer surplus rises, benefiting both insured and uninsured patients. Given their large numbers, Medicare beneficiaries capture the bulk of the increase in surplus. However, as shown in Appendix Table A.8, per patient, the uninsured benefit most from the policy. For example, at a Medicaid-equivalent payment rate, charity patient surplus increases by \$1,235 per patient, Medicaid beneficiaries by \$617 per patient, Medicare beneficiaries by \$467 per patient, and privately insured by \$486 per patient.

These changes, however, come at high fiscal costs, as shown in Figure 7b. For example, at the Medicaid-equivalent level, the program spends \$132 million per quarter. Annualized, this cost is 1.24 times Florida's combined state and federal Medicaid DSH expenditures in federal

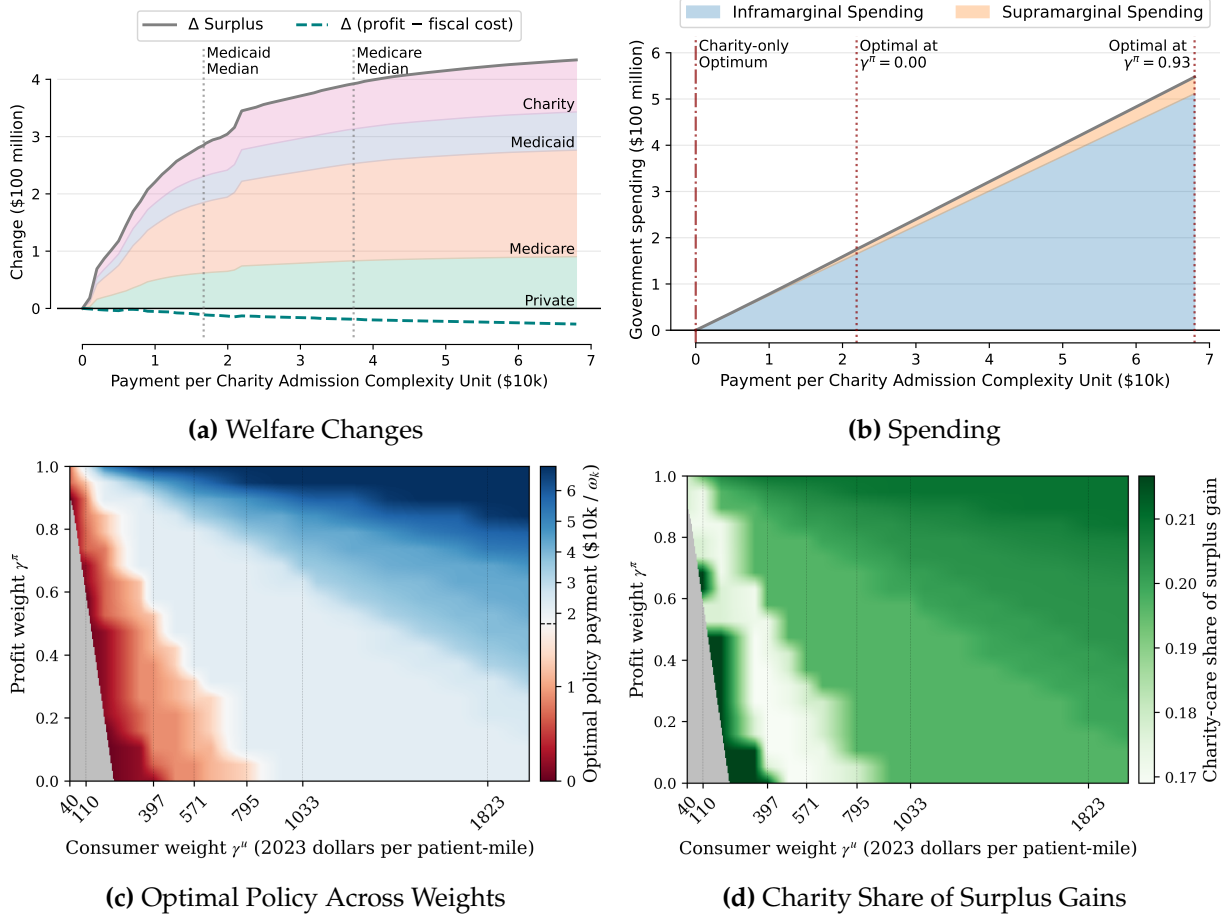


Figure 7: Pay-per-Complexity Policies

Notes: These figures show the equilibrium welfare effects of compensating hospitals for charitable care. Payments are measured in tens of thousands of dollars per DRG-weight of a charity admission. The vertical lines in Figure 7a mark the rates that would match the median payments Medicaid and Medicare give hospitals per DRG-weight unit. In Figure 7b, the dotted vertical lines show the smallest γ^π that would make that payment level optimal, and the dash-dotted line marks the optimum when the regulator values only charity-patient surplus. Supramarginal spending equals total policy payments for additional charity complexity units admitted to within-market hospitals. All remaining spending is inframarginal. Figure 7c shows the optimal policy at a range of γ^π and γ^μ . The color scheme is anchored at the median Medicaid rate: blue colors are at or above the Medicaid level, and red colors are below. The gray area marks where the optimal policy is null. The figure marks estimates of patients' dollar-valued disutility per mile traveled, converted to 2023 dollars and discussed in Appendix E.1. The markers at \$1,033 and \$1,823 are from Prager (2020); those at \$397 and \$795 are from Gowrisankaran et al. (2015), while those at \$110 and \$571 are from Ho and Pakes (2014). We caveat that Ho and Pakes (2014) estimate patient-provider responsiveness to insurer prices rather than patient prices. Finally, Figure 7d shows the share of the patient surplus gain going to charitable care patients in each solution.

fiscal year 2019, or 1.76% of its total Medicaid expenditures (Centers for Medicare & Medicaid Services, n.d.; U.S. Bureau of Labor Statistics, n.d.). Moreover, at any meaningful payment level, government spending is between 2 and 7 times the surplus it delivers to the uninsured. The source of this wedge becomes clear when we separate spending into inframarginal care—the volume of uninsured care delivered without the policy—and the incremental supramarginal care captured from the outside option. As shown, most spending compensates hospitals for care that would have been delivered regardless. Thus, the surplus gains perceived by the uninsured in Figure 7a come from improved sorting rather than from extensive-margin gains. An immediate consequence of

this wedge is that, if we ignore other factors, the optimal policy is to provide hospitals with no additional compensation. This finding aligns with evidence that recipients can value subsidized insurance for low-income adults below its gross fiscal cost (Finkelstein et al., 2018, 2019a,b) and points to the inframarginality of the care delivered as a source of this wedge.

This perspective, however, is overturned once the benefits to non-recipients are incorporated into the analysis. Figure 7b and Table 3 show that the surplus-optimal policy level—ignoring any benefit to hospitals—pays hospitals 31.4% over Medicaid, or about 58.7% of Medicare rates. Total patient surplus increases by \$345.0 million, with only 19.6% of that benefit going to the uninsured. At that level, hospital profits increase by \$161.9 million, and government spending by \$174.9 million. Moreover, payment levels from this optimum to the median private rate (the upper bound in the figure) can be rationalized by weights on hospital profits ranging from 0 to 0.93 relative to fiscal cost. Relative to the surplus-optimal level, the upper bound on the policy pays hospitals rates equivalent to those paid by private insurers, tripling government spending and hospital profits while increasing patient surplus gains by only 25.7%. Payment at Medicare levels increases government spending by 71.2% relative to the surplus-optimal level, and increases surplus gains by 13.7%.

Table 3 also shows the effect of these policies on available hospital capacity, which equals allocated resources times each line’s acceptance rate. Pseudo-insuring the uninsured increases emergency and scheduled available capacity by hundreds of thousands of bed-days. In contrast, this reallocation negatively affects insured patients seeking scheduled care; at the Medicaid-equivalent policy, it removes access to 5,358 bed-days for these patients. However, the increase in access outweighs the loss in scheduled insured care, so total idle capacity decreases by 7.46 percentage points under the Medicaid-equivalent policy.

The optimal payment rate depends on the regulator’s relative preferences for patient surplus relative to fiscal spending. Three findings, however, hold across a wide range of preferences. First, the optimal policy compensates hospitals for charitable care at rates comparable to Medicaid. Figure 7c shows the optimal policy as the regulator’s preference varies across a range of estimates from the literature. In 4.6% of the graph (gray area), the optimal policy is inaction, while in 16.9% (red area), the optimal policy is to pay hospitals at rates below Medicaid rates. In all other configurations (blue), the optimal policy is at or above Medicaid levels, making it the dominant outcome. Second, as Figure 7d shows, most surplus gains from the optimal policy accrue to non-beneficiaries. In particular, the uninsured obtain only 19.7% of the average patient surplus gain, and never more than a quarter of the total. Finally, a policy analysis centered exclusively on the value of insurance to the uninsured will recommend inaction or very minimal payments (see Appendix Figure A.12)—resource externalities make the policy worth its cost.

Table 3: Counterfactual Policy Welfare Effects

	Regulatory	Δ Welfare (\$100M)				Δ Available Capacity (1k bed-days)			
	Price	Patients		Hospitals	Gov.	Emergency	Scheduled		Idle
	z^u, z^p	All	Charity				Insured	Uninsured	(p.p.)
Panel A: Pay-per-Complexity									
Medicaid-equivalent	1.668	2.849	0.545	1.214	1.320	155.423	-5.358	103.931	-7.460
Medicare-equivalent	3.732	3.923	0.789	2.807	2.995	213.085	-4.997	175.472	-11.265
Optimum at $\gamma^\pi = 0$	2.192	3.450	0.678	1.619	1.749	191.112	-2.463	145.155	-9.803
Optimum at $\gamma^\pi = 0.93$	6.797	4.339	0.908	5.203	5.476	230.867	-20.289	185.055	-11.619
Panel B: Floor-and-Trade (10% tax-at-risk)									
Unconstrained	11.686	2.516	0.545	-0.172	0.000	119.247	-19.013	110.235	-6.181
Homogeneous	0.251	0.310	0.141	-0.002	0.000	13.831	-5.085	25.268	-0.999

Notes: The first column reports the regulatory payment or credit price in \$10,000 units. Welfare is measured as the change relative to the baseline, in hundreds of millions of dollars. Patient columns report gains for all patients and charitable-care patients, respectively. “Gov.” represents the change in fiscal cost. Available capacity is the sum across hospitals of the product of each line’s allocated bed-days and its acceptance rate, measured in thousands. The idle capacity ratio is defined as $(\sum_h B_h - \sum_{h,j} a_{hj} b_{hj}) / \sum_h B_h$, and the change is presented in percentage points. For comparison, our counterfactual analysis includes 133 hospitals, with a total capacity of 3,404,941 bed-days in 2019Q4. All values are quarterly.

8.1.1 Competitive Effects. The analysis of resource externalities and their interaction with payment policies might suggest that most underlying frictions can be captured in a single-firm model. The equilibrium, however, plays a subtle yet important role. Beyond allowing us to rationalize the full vector of patient acceptance probabilities, competitive interactions between hospitals have an important yet ex-ante ambiguous effect on the policy’s effectiveness. For example, if the policy makes one hospital increase its emergency care availability at the expense of scheduled insured care, this would translate into higher demand for insured scheduled care at rivals and thus a greater incentive for them to reallocate capacity towards that care. This, in turn, would make it more expensive for the regulator to incentivize those hospitals to provide more emergency care. Alternatively, increased emergency care availability at one hospital might reduce charitable care demand at its rivals, increasing their incentives to provide that care and lowering the policy’s cost.

To resolve this ambiguity, we compute each hospital’s best response to the policy while holding their rivals’ actions at the status quo. The difference between equilibrium and these direct effects reveals the competitive effect, which, as we show in Appendix E.3, substantially affects market outcomes. In particular, competitive effects lead hospitals to allocate more resources to scheduled insured care and fewer to emergency care, consistent with their increased incentives to capture profitable displaced patients. It also significantly reduces idle capacity and modestly increases the policy’s fiscal cost. Crucially, competitive effects significantly boost the policy’s surplus gains: at the Medicaid-equivalent level, competitive effects account for almost a third of the patient surplus gains, with most incremental gains accruing to the insured. Intuitively, rivals’ response to reallocated insured demand helps mitigate the negative effects of resource externalities.

8.1.2 Medicaid Expansions. These pseudo-insurance simulations vary only hospital compensation, which differs from what a true insurance expansion would entail. In particular, a Medicaid expansion would change patients' networks, the payment structure to hospitals, the rate at which charity patients require emergency care relative to scheduled care, and the service lines they use.³⁹ To assess the impact of these changes, we simulate their outcomes and present the results in Appendix Table A.9. We find that the true Medicaid rate structure would increase patient surplus by an additional \$9.4 million relative to the Medicaid-equivalent pseudo-insurance, with \$4.6 million of that increase going to charity-care patients. Holding charity care patient aggregate risk constant but adjusting their emergency-to-scheduled illness ratio to match that of Medicaid beneficiaries of similar age and gender, patient surplus gains would nearly double to 542.8 million. Importantly, at this point the uninsured would receive 59.4% of the policy's total surplus gain—enough to justify the fiscal cost even ignoring benefits to the insured. From this point, restricting patients to Medicaid networks and moving them to the same service line as other patients slightly decreases their surplus gains by about 1.3%, resulting in a net increase of \$535.4 million per quarter, with 59.8% of the gains falling on the formerly uninsured. These results suggest that a true Medicaid expansion would not overturn the analysis above.⁴⁰

8.2 Floor-and-Trade Policies

We now turn to policies that leverage hospitals' non-profit tax exemption to incentivize charitable care. Congress and health-policy circles have proposed minimum requirements on non-profit hospitals' charity-care or community-benefit obligations (U.S. Senate Committee on Finance, Minority Staff, 2007; U.S. Senate Committee on Health, Education, Labor, and Pensions, Majority Staff, 2023; Wofford et al., 2024). The topic has drawn interest from politicians across the aisle, with proposals from Republican Senator Charles E. Grassley and Independent Senator Bernie Sanders recommending variants of tying hospital tax benefits to their performance, drawing on a similar policy in Texas (Kennedy et al., 2010).⁴¹ However, a natural concern is that under binding minimum standards, many hospitals would be unable to provide the required care because demand

³⁹The ratio of emergency to scheduled care will likely change as the data reflect uninsured patients' response to the low availability of scheduled charitable care.

⁴⁰Most of the surplus increase from expanding Medicaid comes from how it reshapes the composition of emergency to scheduled care for charitable care patients. We note that here, emergency care usage implies an admission and is distinct from evidence on ED emergency use increases following coverage expansions. Prior evidence on this margin from insurance expansions is mixed, with some finding statistically insignificant effects and others documenting decreases (Finkelstein et al., 2012; Kolstad and Kowalski, 2012; Finkelstein et al., 2019a; Garthwaite et al., 2019).

⁴¹Texas imposes minimum requirements on the community benefits provided by non-profit hospitals to qualify for state and local tax exemptions (State of Texas, 2026a; Texas Department of State Health Services, 2025; State of Texas, 2026b). Hospitals must either provide charity and government-sponsored indigent care at a cost equal to the hospital's tax-exempt benefits or provide community benefits of at least 5 percent of net patient revenue, with at least 4 percent for charity and indigent care. Hospitals, however, may also appeal to a needs-based standard to justify that their provision of charitable and indigent care is reasonable relative to their community needs and the hospital's available resources.

is not present or because their financial situation would force them to shut down. An alternative that alleviates this concern is to implement a floor-and-trade system, as proposed in academic circles by Dranove et al. (2015). Under this system, non-profit hospitals must provide a minimum level of charitable care, and a market is created for hospitals to buy and sell charitable-care credits, allowing those with little uninsured demand to finance those with a significant uninsured burden. Such policies have subsequently been discussed in congressional testimony and state legislative deliberations (Levinson, 2023; Georgia House of Representatives, 2019). Importantly, these policies require no additional governmental spending; instead, they demand larger market interventions.

We focus on two variants of floor-and-trade policies that differ in their regulatory discretion. First, we consider an *unconstrained* policy where the regulator can freely allocate obligations and credits across hospitals, allowing it to adapt to each hospital's setting and financial situation. Second, we consider a *homogeneous* policy that must give each hospital the same obligation as a share of its profits. To present a conservative perspective on the potential of these policies and to recognize that non-profit hospitals might provide additional value to the community beyond charitable care, we require our floor-and-trade policies to be incentive-compatible with all existing non-profit hospitals retaining their status. For this reason, we also tie the homogeneous policy floor to profits rather than revenue. In our sample, any meaningful requirement on non-profit supply relative to revenue or expenses would cause some of the largest suppliers of charitable care to drop their non-profit status when the credit price is too low, highlighting the challenges created by a minimum-requirement policy with no trade. We present our main analysis assuming a moderate 10% tax at risk for hospitals, and hold γ^u to our preferred value of \$1,033 per mile and $\gamma^\pi = 1$ to minimize the policy's negative effect on hospitals.⁴²

The unconstrained policy consists of two elements: a vector of endowments $\mathbf{u}$, whose components are each hospital's credit or obligation in charity-care complexity units; and z^p , the equilibrium trading price. Thus, the policy imposes a fixed participation cost $u_h z^p$ on each hospital and sets a payment z^p per unit of charitable care. As these payments mirror those under pseudo-insurance, non-profit hospitals in the floor-and-trade system behave as in the analysis of the previous section. The main distinction is that, while in the previous section the regulator set the price, here z^p is an equilibrium object which depends on the regulator's chosen endowment vector. In particular, conditional on $\mathbf{u}$, the market-clearing condition determines the trading price, while $\mathbf{u}$ maximizes total welfare subject to incentive compatibility constraints. These two conditions are:

$$\sum_{h \in \mathcal{H}^{np}} \left(\sum_{j \in \mathcal{J}_h} \sum_{k \in \mathcal{K}_h^u} \omega_k a_{hj} \lambda_k R_{hk}(\mathbf{a}) - u_h \right) = 0 \quad (18)$$

$$(1 - \gamma^{\text{tax}}) \sum_{j \in \mathcal{J}_h} \pi_{hj}(a_{hj}^{fp}, b_{hj}^{fp}; p^c) \leq \sum_{j \in \mathcal{J}_h} \pi_{hj}(a_{hj}, b_{hj}; p^c + z^p) - z^p u_h \quad \forall h \in \mathcal{H}^{np} \quad (19)$$

⁴²Appendix Table A.10 shows the outcomes of increasing the tax at risk.

where $\mathcal{H}^{np}$ is the set of non-profit hospitals, firm actions (a_{hj}, b_{hj}) and $(a_{hj}^{fp}, b_{hj}^{fp})$ are optimal given (z^p, u) , with the latter denoting those taken if the hospital became a for-profit, with no access to the charity care trading system and facing a tax γ^{tax} on profit.⁴³ Here we denote $\pi_{hj}(a, b; p^c)$ as the hospital-line expected profit function, taking an extra argument for the payment per uninsured admission complexity unit, where p^c is the status-quo payment rate of charitable care admissions. The following proposition, proven in Appendix E.2, provides an important property of this policy.

Proposition 2. *The unconstrained floor-and-trade market clearing and incentive compatibility conditions hold if and only if total participation profit is greater than or equal to the total value of credit obligations:*

$$\sum_{h \in \mathcal{H}^{np}} \sum_{j \in \mathcal{J}_h} \left(\pi_{hj}(a_{hj}, b_{hj}; p^c + z^p) - (1 - \gamma^{tax}) \pi_{hj}(a_{hj}^{fp}, b_{hj}^{fp}; p^c) \right) \geq z^p \left(\sum_{h \in \mathcal{H}^{np}} \sum_{j \in \mathcal{J}_h} \sum_{k \in \mathcal{K}_j^u} a_{hj} \lambda_k R_{hk} \omega_k \right). \quad (20)$$

This proposition has two benefits. First, it lets us replace $H^{np} + 1$ constraints in the planner's problem with a single scalar constraint. Second, it allows us to switch the planner's control variable from the endowment vector to the trading price: for any credit price satisfying the condition of Equation (20) and any vector of hospital actions that are individually optimal under this price, there is an endowment vector that makes the outcome an equilibrium. Thus, the planner problem is to maximize $W(z; \gamma^u, \gamma^\pi)$ as in Equation (17), subject to the aggregate constraint.⁴⁴

The first row of Panel B in Table 3 shows the impact of the optimal unconstrained floor-and-trade policy. It delivers 88.3% of the surplus gains of the Medicaid-equivalent pseudo-insurance policy, with an almost identical gain for charity patients. Thus, floor-and-trade is comparatively more targeted toward the uninsured, with 21.6% rather than 19.1% of the consumer surplus gain accruing to them. The policy also reduces hospital profits by \$17.2 million, increases available emergency care capacity slightly less than Medicaid-equivalent pseudo-insurance does, and skews scheduled care slightly more toward charity. In comparable units, idle capacity also decreases, with a 6.2 percentage-point gain in total available capacity.

The policy's trade patterns show where charitable care supply and demand are mismatched, highlighting substantial geographic dispersion in hospitals' trading positions, as shown in Appendix Figure A.13. For example, AdventHealth Orlando is the largest buyer of credits in this analysis, buying 23.0% of all traded credits for a total of \$2.72 million. At the same time, Holy Cross Hospital in Fort Lauderdale is the largest seller, supplying 22.5% of all credits at \$2.66 million. These two hospitals also highlight the policy's benefit, as it enables credits to flow between

⁴³In this analysis, we assume that the hospital's preferences for uninsured care are not a feature of their non-profit status, but rather organizational goals. Relaxing this assumption makes for-profit behavior worse for hospitals, which relaxes the incentive compatibility constraint, and strengthens the policy. This is because most charity care is delivered through the emergency line, which for-profit hospitals cannot avoid without also avoiding profitable emergency patients.

⁴⁴Note that the objective of the planner remains the same as in the pseudo-insurance problem because credit trade costs and revenues cancel out in aggregate. The policy's fiscal cost is zero by construction.

hospitals located 182 miles apart—a distance very few patients are likely to cross for care.

The heterogeneity in local uninsured demand does not fully explain the mismatch between the supply and demand of charitable care. For example, the second-largest buyer of credits is in Palm Beach County, which has a higher uninsured rate than Broward County, where Fort Lauderdale and the largest seller of credits are located. In particular, the correlation between a hospital's optimal net credit position and its local uninsured rate is -0.19 , indicating that higher-burden hospitals sell more credits, as expected. Thus, while evidence and theory agree that the largest credit buyers should be those with the largest tax benefits relative to uninsurance burden, it is hard to predict which hospitals are in the right position to support, or be supported by, the system. The flexibility of the unconstrained policy allows the regulator to target credit obligations based on hospitals' incentive-compatibility constraints and to maximize valuable trade. The outcome of the homogeneous policy illustrates this value best, as we discuss next.

Unlike the unconstrained policy, homogeneous floor-and-trade must find a common target for all hospitals. This homogeneity breaks the result of Proposition 2, forcing the planner to solve the program implied by the full set of incentive-compatibility constraints and market-clearing conditions.⁴⁵ The second row of Panel B in Table 3 shows that although the homogeneous policy improves patient surplus overall, it achieves only 12.3% of the surplus gain of the unconstrained policy. This stark drop highlights the cost of trying to fit the vast heterogeneity in patient demand, competition, and hospital resources with a single target based on hospitals' finances. Appendix Figure A.13 shows that trading positions under this homogeneous policy differ from those in the unconstrained case and have only a mild positive correlation of 0.24. For example, while the largest seller of credits in the unconstrained regime remains a seller under the homogeneous policy, the largest buyer of credits becomes a net seller—the homogeneous rule cannot raise obligations for some hospitals without violating other hospitals' incentive-compatibility constraints.

8.3 Discussion

The counterfactual policy analysis shows how limited insurance access for some affects all who share the same resources, regardless of insurance. Hospitals facing higher uninsured demand limit their exposure to payment risk by reallocating resources from emergency to scheduled care. This reallocation forces patients to seek care in less preferred locations, reduces access to local emergency care, and leaves capacity idle. These effects occur despite hospitals placing considerable value on uninsured care; thus, they are not an outcome of pure profit-seeking behavior.

These negative effects can be mitigated by compensating hospitals for uninsured care, but the uninsured would likely not value this policy more than it would cost to run. However, the externalities produced by resource reallocation would benefit the insured substantially, providing

⁴⁵The policy's homogeneity keeps it a scalar problem. The equilibrium trading price is determined by the fixed point of the market-clearing condition.

a stronger rationale for such policies. Moreover, the analysis of floor-and-trade policies shows that a large fraction of the consumer surplus gains from direct compensation policies can be achieved at negligible public cost by targeting precisely where non-profit tax benefits exceed local demand for charitable care and where they fall short. However, the analysis also cautions that this policy requires targeting, as the underlying heterogeneity driving the misalignment between non-profit benefits and burden cannot be easily corrected by a uniform policy.

The gains to the insured from increased access to emergency care raise the question of why private prices do not reflect this value. Intuitively, one might expect private emergency prices to increase to incentivize emergency resource allocation. Three key market factors, however, stand in the way: non-discrimination in emergency care, unobservability of resource allocation, and insurance competition. The first implies that any insurer who increases emergency payments to increase capacity will naturally benefit the enrollees of all other insurers. The second implies that these payments cannot be made contingent on resource allocation. The third implies that any insurer that increases compensation for emergency care would place itself at a disadvantage relative to rivals, giving them better admission rates at lower cost. Thus, much like pseudo-insurance policies, each insurer's attempt to offset the underlying distortion would create positive externalities for its rivals at high cost to itself and with limited reward.

9 Conclusion

This article studies the private and public costs of charitable care. Our empirical strategy exploits cross-state spillovers from Medicaid expansions, channeled through non-profit systems operating in both expansion and non-expansion states. Spillover hospitals in non-expansion states increased their supply of charitable care, shifting demand within those states. Using the resulting reallocation of care at nearby hospitals, we measure the impact of charitable care on hospitals' costs and insured patient access. We combine this variation with a framework of hospital competition for patients via resource allocation and admission control, which we estimate using detailed data from Florida. The model allows us to recover hospitals' cost structure and simulate policies that compensate them for otherwise charitable care. We find that uninsured patients' willingness to pay for the policy falls well short of its fiscal cost, but that its positive effect on access to emergency care for all patients greatly benefits the already insured, compensating for the policy's cost. We find that the patient-optimal pseudo-insurance policy resembles Medicaid, and that a similar outcome can be obtained at negligible fiscal spending under a well-targeted floor-and-trade system.

Several avenues for future research follow. First, we hold hospitals' total capacity and total charitable care demand fixed within a period and abstract from extensive-margin adjustments. Extending the analysis to account for entry and potential consequences for insurance enrollment is an important avenue for future work. Second, our analysis is agnostic about the drivers of hospitals' preference for charitable care. While we offer potential mechanisms, identifying the

origin of this behavior could contribute greatly to our understanding of hospital operations. Finally, our framework abstracts from dynamic patient reallocation and the impacts of congestion on quality and wait times. Identifying and estimating a comprehensive framework would likely require significant additional data and variation. Still, it could further our understanding of the externalities of uninsurance and the potential need to expand hospital resources. We view each of these extensions as complementary to, rather than in tension with, the article’s central finding that the costs of charitable care are borne well beyond the uninsured, and that evaluating policies only by what they deliver to their intended beneficiaries will systematically undervalue them.

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